



The Inertial Labs KERNEL IMU

Interface Control Document

Models:

KERNEL-100
KERNEL-102
KERNEL-108
KERNEL-110
KERNEL-110M
KERNEL-111
KERNEL-120

KERNEL-201
KERNEL-210
KERNEL-220

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1. Scope and Applicability

This Interface Control Document (ICD) provides details on mechanical mounting, the electrical connections, powering and communication between the Inertial Labs KERNEL IMU and a host computer or system. This document is intended for all parties requiring such information, including engineers and researchers responsible for implementing the interface.

2. Introduction

The Inertial Labs KERNEL Inertial Measurement Unit (IMU) is an Advanced MEMS-based sensor that is a compact and self-contained strapdown inertial measurement system. This device measures absolute pitch, and roll angles, linear accelerations and angular rates using three-axis high-grade MEMS accelerometers and three-axis MEMS gyroscopes. Angular rates and accelerations are determined with high accuracy for both static and dynamic applications.

All KERNEL IMU sensors are precision-aligned across axes and calibrated over their operational temperature ranges.

Figure 2.1 shows the KERNEL IMU and its own coordinate system, Oxyz. This coordinate system is body-fixed and defined as the calibrated sensors coordinate system. Non-orthogonality between axes of the body-fixed coordinate system is an order of 0.01° .

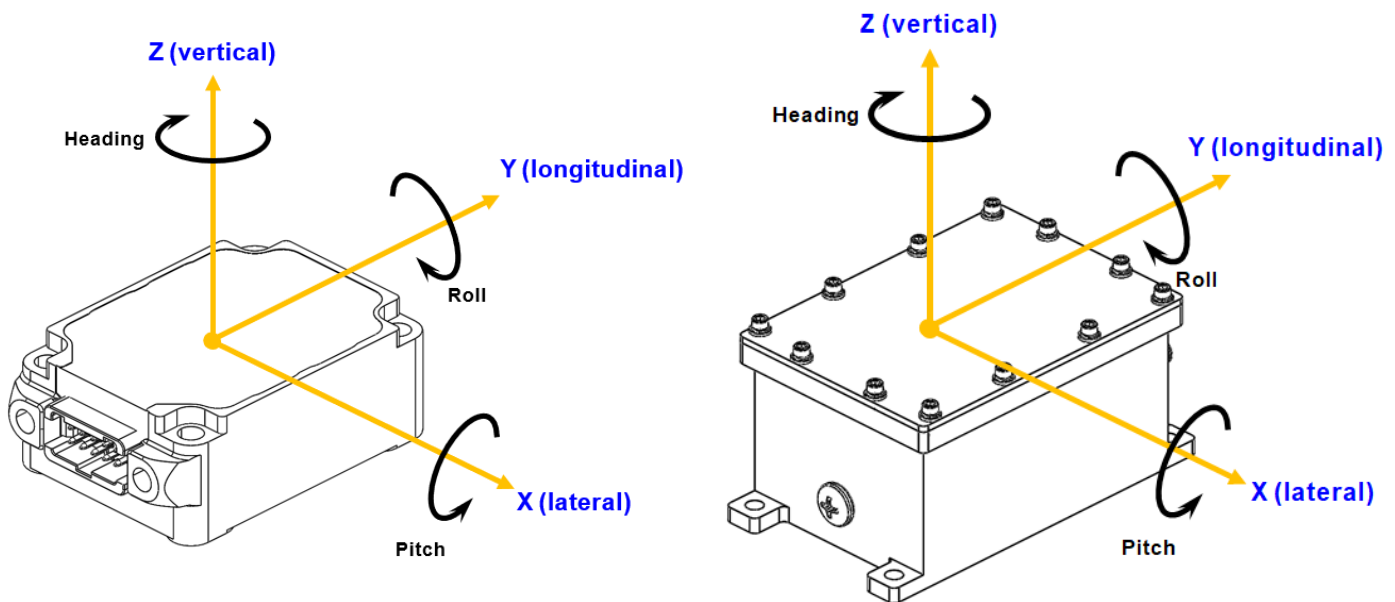


Figure 2.1. The KERNEL IMU Coordinate System

Inertial Labs provides the following models of KERNEL IMU (see Figure 2.2):

- KERNEL-100 – consists of a three-axis MEMS accelerometer and three-axis MEMS gyroscope. The measured linear accelerations and angular rates are being processed on the first generation of the Inertial Labs processing board.
- KERNEL-110, KERNEL-111, KERNEL-108 – consist of a three-axis MEMS accelerometer and three-axis MEMS gyroscope. The measured

linear accelerations and angular rates are being processed on the second generation of the Inertial Labs processing board.

- KERNEL-120 – consists of a three-axis MEMS accelerometer, three-axis MEMS gyroscope, and additional high range three-axis MEMS accelerometer. The measured linear accelerations and angular rates are being processed on the second generation of the Inertial Labs processing board.
- KERNEL-201 – consists of industrial-grade three-axis MEMS accelerometer, three-axis MEMS gyroscope, and a high-range three-axis gyroscope as an option. Member of the third generation of the Inertial Labs KERNEL IMU family.
- KERNEL-210 – consists of tactical-grade inertial sensors such as a three-axis MEMS accelerometer and three-axis MEMS gyroscope. Member of the third generation of the Inertial Labs KERNEL IMU family.
- KERNEL-220 – consists of tactical-grade inertial sensors such as a three-axis MEMS accelerometer, three-axis MEMS gyroscope, and additional high range three-axis MEMS accelerometer. Member of the third generation of the Inertial Labs KERNEL IMU family.
- KERNEL-110M – KERNEL-110 enclosed in case with MIL-STD-1275 protection.

Important note: In this document, all information related to KERNEL-100 is correct for KERNEL-110, KERNEL-111, KERNEL-108, KERNEL-120, KERNEL-201, KERNEL-210, and KERNEL-220 unless otherwise specified. Also, all information related to the KERNEL-110 is correct for the KERNEL-108, KERNEL-111, and KERNEL-110M unless otherwise specified.

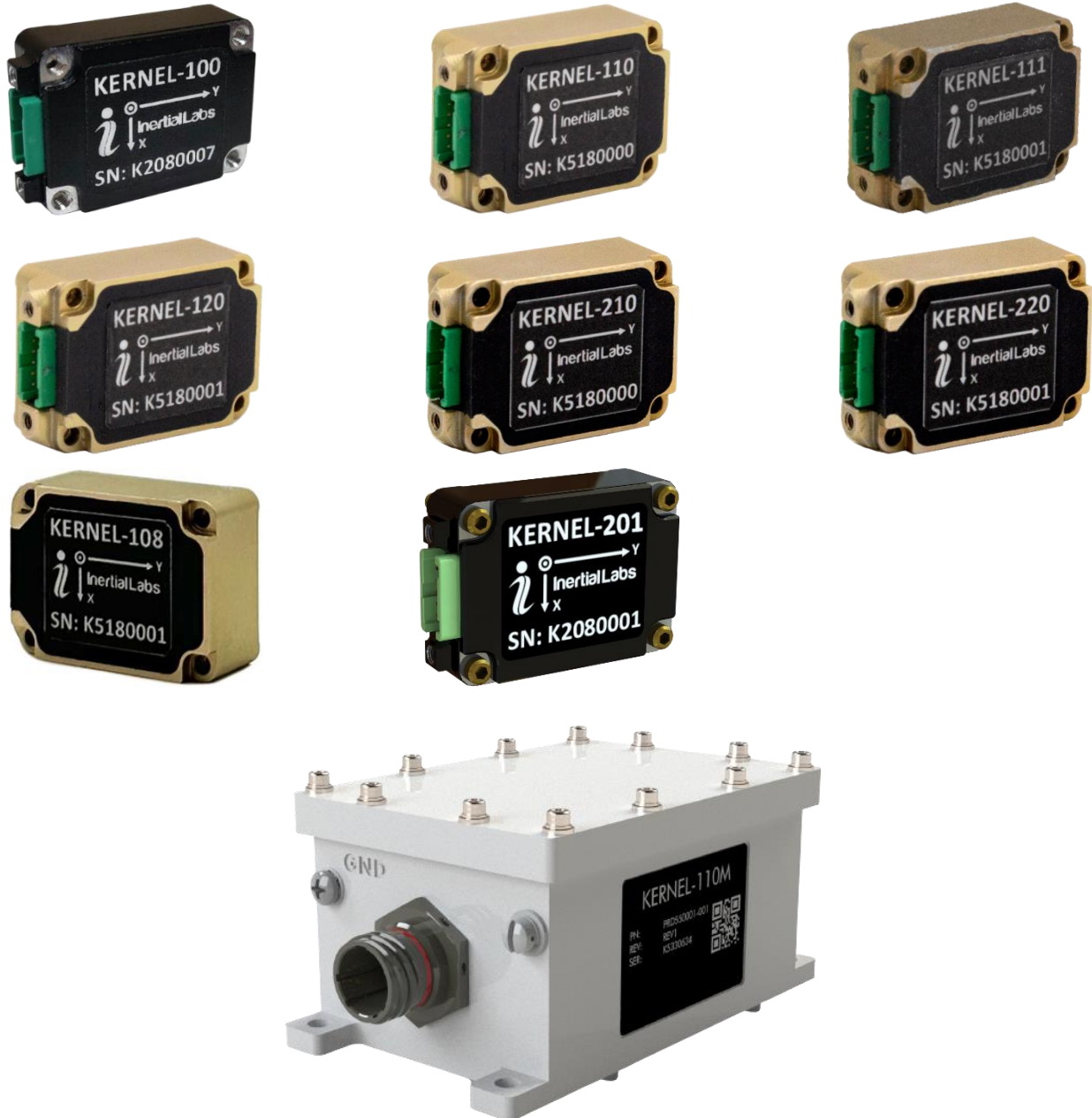


Figure 2.2. The Inertial Labs KERNEL IMU models

3. Specifications

See the datasheet intended for the certain Inertial Labs KERNEL IMU.

4. Mechanical Interface

4.1. KERNEL-100 and KERNEL-102

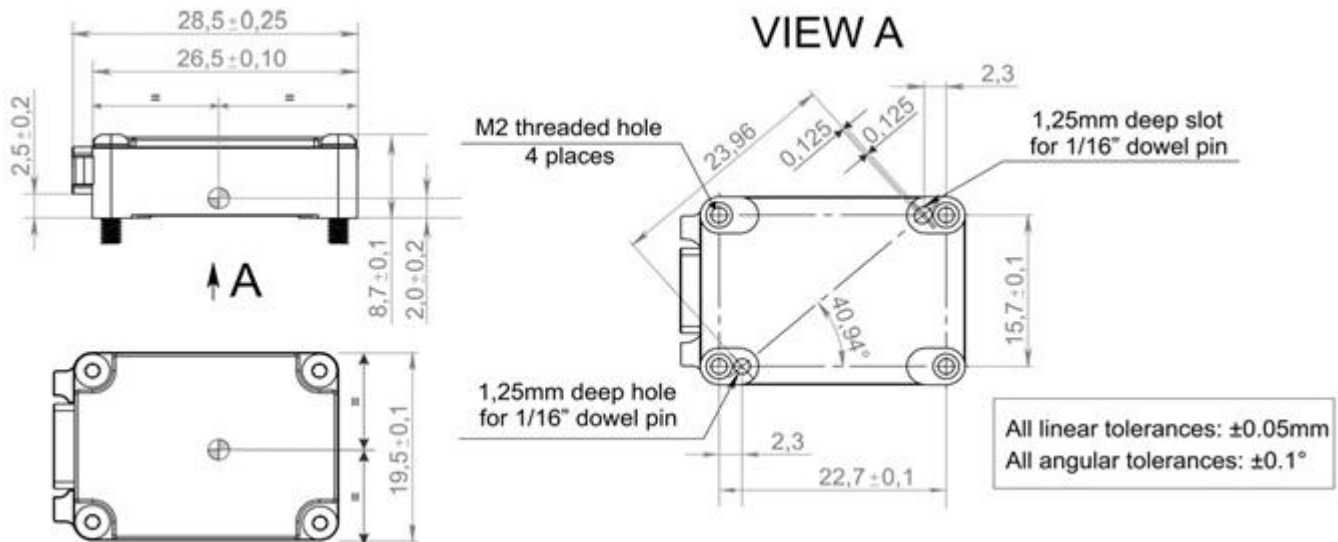


Figure 4.1. The Inertial Labs KERNEL-100 and KERNEL-102 outline drawing

Notes:

1. All dimensions are in millimeters (mm).
2. Tolerances:
 - .x ± 0.05 mm
 - .xx ± 0.02 mm
3. Surface finish: Type III Class 2 hard anodic coating per MIL-A-8625 (black color).

4.1.1. Mechanical Mounting

The Inertial Labs KERNEL IMU should be mounted on your system as shown in Figure 4.2 using mounting option 1 or 2 as denoted in the drawing.

Recommended requirements for this mounting surface of the carrier object are as follows: flatness tolerance of 0.03mm; undulation of Ra=1.25.

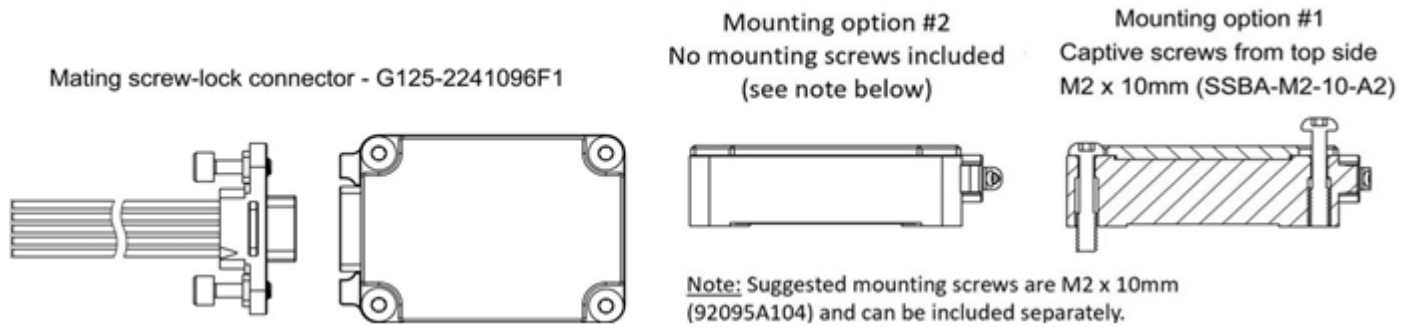


Figure 4.2. Mounting options for the KERNEL-100.

4.2. KERNEL-108, KERNEL-110, KERNEL-111, KERNEL-120, KERNEL-201, KERNEL-210, and KERNEL-220

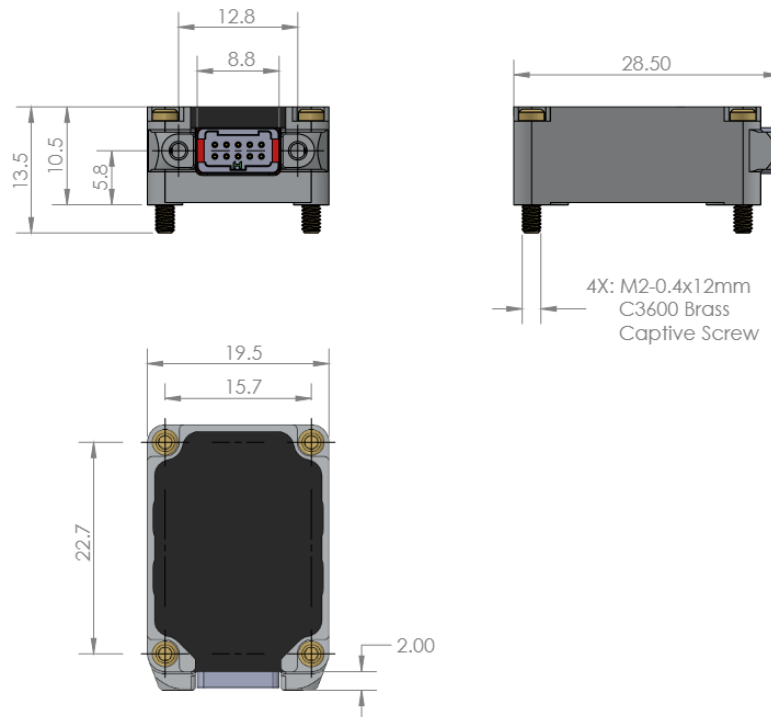


Figure 4.3. The Inertial Labs KERNEL-108, KERNEL-110, KERNEL-111, KERNEL-120, KERNEL-210, KERNEL-220, and KERNEL-201 outline drawing

Notes:

1. All dimensions are in millimeters (mm).
2. One place decimal tolerance: ± 0.25 .
3. Weight: 10.2 $\pm$ 0.25 g wo screws, 11.0 $\pm$ 0.25 with screws.
4. Finish: abrasive blasting (#120-#150 grit).
5. Surface Treatment: conversion coating per MIL-DTL-5541 Type I.

4.2.1. Center of Navigation

The position of the center of navigation depends on the specific product type. Choose the needed product code from the table below to see the list of applicable drawings.

Product code	Applicable figures
KERNEL-120-G2000-A40A90-TGA-C12-S-V1.2	Figure 4.4, Figure 4.5, Figure 4.6, Figure 4.7, Figure 4.8
KERNEL-120-G2000-A15A90-TGA-C12-S-V1.2	Figure 4.4, Figure 4.5, Figure 4.6, Figure 4.7, Figure 4.8
KERNEL-120-G2000-A8A90-TGA-C12-S-V1.2	Figure 4.4, Figure 4.5, Figure 4.6, Figure 4.7, Figure 4.8
KERNEL-108-G2000-A40-TGA-C12-A-V1.2 KERNEL-110-G2000-A40-TGA-C12-A-V1.2 KERNEL-111-G2000-A40-TGA-C12-A-V1.2	Figure 4.4, Figure 4.5, Figure 4.6, Figure 4.8
KERNEL-108-G2000-A15-TGA-C12-A-V1.2 KERNEL-110-G2000-A15-TGA-C12-A-V1.2 KERNEL-111-G2000-A15-TGA-C12-A-V1.2	Figure 4.4, Figure 4.5, Figure 4.6, Figure 4.8
KERNEL-108-G2000-A8-TGA-C12-A-V1.2 KERNEL-110-G2000-A8-TGA-C12-A-V1.2 KERNEL-111-G2000-A8-TGA-C12-A-V1.2	Figure 4.4, Figure 4.5, Figure 4.6, Figure 4.8

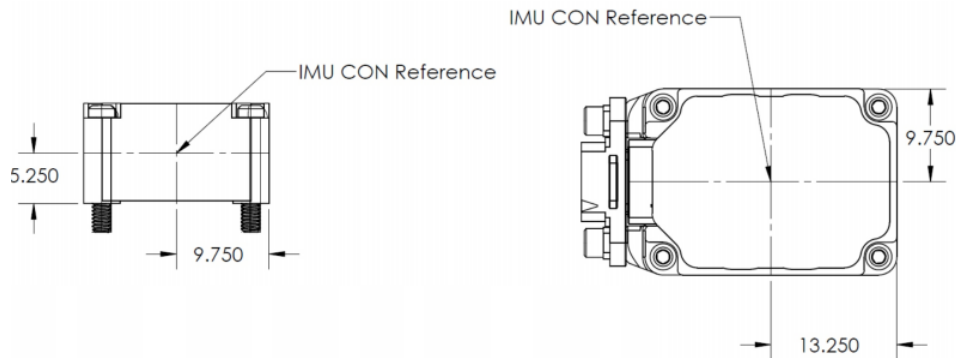


Figure 4.4. IMU housing center of navigation reference

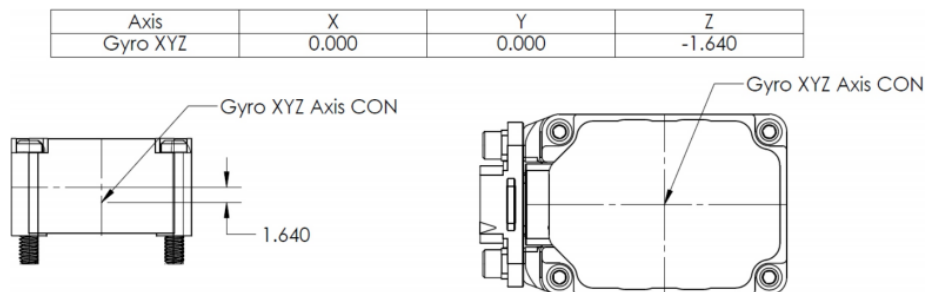


Figure 4.5. Gyro center of navigation reference

Axis	X	Y	Z
XY	0.000	5.230	-0.120
Z	0.000	10.460	-0.570

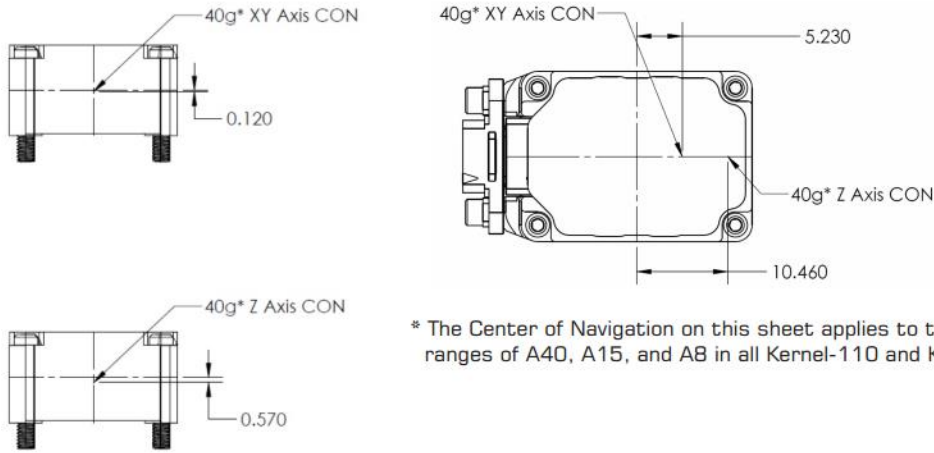


Figure 4.6. Accel 40g center of navigation reference

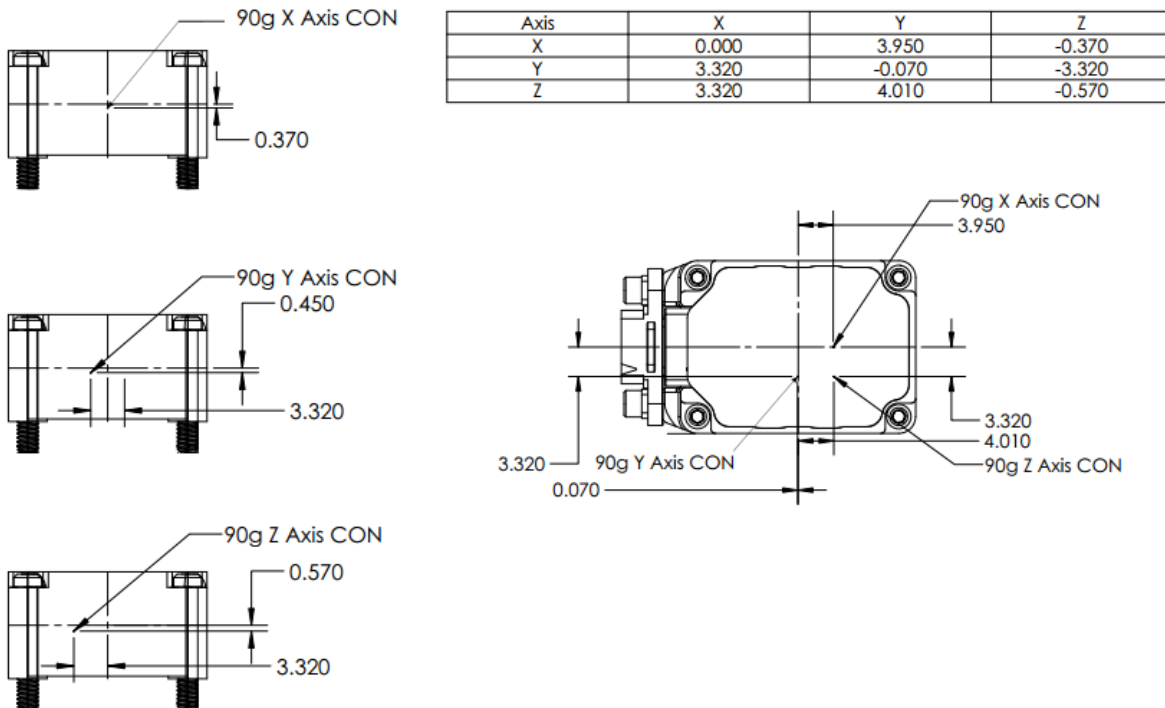


Figure 4.7. Accel 90g center of navigation reference

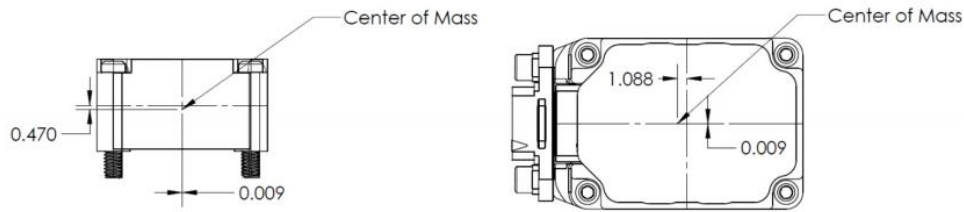


Figure 4.8. IMU center of mass with interface connector

4.3. KERNEL-110M

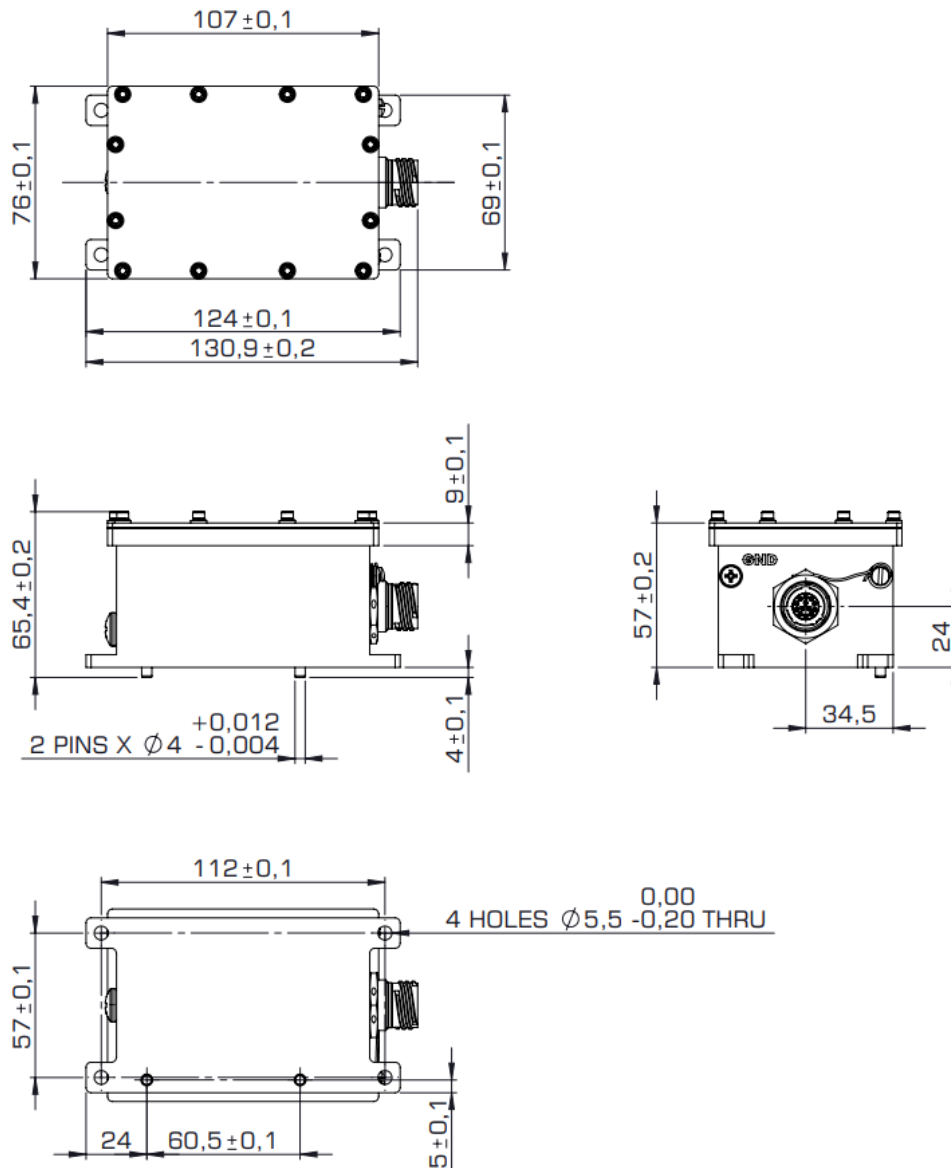


Figure 4.9. The Inertial Labs KERNEL-110M outline drawing

Enclosure Painting & Engraving Specification:

1. White color, code #27885, according to FEDSTD-595
2. Primer. One layer of lead and chromate-free Epoxy Primer according to MIL-DTL-53022 and manufacturer's instructions. Layer thickness 25-40µm. Drying time before intermediate application: 8-24 hours at room temperature conditions.
3. Inter Mediate. Two layers of Epoxy Polyamide according to MIL-PRF-22750 and manufacturer's instructions. Each layer thickness: 30-40µm, White Color #27885 according to FED- STD-595. Drying time between layers and topcoat application: 6-48 hours at room temperature conditions.
4. Topcoat. Two layers of Polyurethane according to MIL-PRF-85285 and manufacturer's instructions. Each layer thickness: 30-40µm. White Color #27885 according to FED- STD-595. Drying time between layers: 1-24 hours at room temperature conditions.
5. Total Thickness. The total dry film thickness of the whole paint system is 145-200µm.
6. Label. Letters filled with Polyurethane according to MIL-DTL-53039. Black Color #27038 according to FEDSTD-595.

5. Electrical Interface

5.1. Standard Configuration

The Inertial Labs KERNEL IMU uses the Harwin 10-pin connector, part number G125-MV11005L0P (screw-lock), for electrical connection to the host system.

The host system should have a cable with the appropriate mating connector – a Harwin female 10 pin cable, part number G125-2241096F1 (mating option #2), shown in Figure 4.2.

Figure 5.1 shows the 10-pin connector pinout on the Inertial Labs KERNEL IMU. Table 5.1 contains the pin diagram of this connector and appropriate pin descriptions of the wires in the 10-pin cable part number G125-MV11005L0P by Harwin.

Table 5.1. Hardware layer properties

1	POWER	Power Supply Input
2	BOOT	FW Update Pin
3	1PPS	1PPS input
4	RS422-A	RS-422 Non-Inverting Input
5	RS422-B	RS422 Inverting Input
6	GROUND	Power Supply Return
7	TOV	Time of Validity Output
8	EXTRIG	External Trigger Input
9	RS422-Y	RS-422 Non-Inverting Output
10	RS422-Z	RS422 Inverting Output

Note: Only KERNEL-100, KERNEL-108, KERNEL-110, KERNEL-120, KERNEL-210, and KERNEL-220 devices support the EXTRIG function.



Mating Option #2

Figure 5.1.

Notes:

- 1PPS, TOV and EXTRIG signals are 3V3 TTL and not 5V tolerant. All pins comply with the IEC 61000-4-2 level 4 standard of ESD protection. Electrical specifications can be seen below in Table 5.2. The External trigger pin can be used to signal the device to output data after the signal is received from a host computer.
- For KERNEL-100 – BOOT pin has internal 10K pull-up resistor to 3.3V. Leave floating if not used. KERNEL-100 enters a firmware update state if BOOT pin is held low when supply power is applied.
- For KERNEL-108/110/120/210/220 – BOOT pin has internal 10K pull-down resistor to supply return. Leave floating if not used. Device enters a firmware update state if BOOT pin is held at logic high (3.3V) when supply power is applied. If driving voltage is higher than 3.3V use a resistive divider as Figure 5.2 shows.
- 1PPS signal resets the time field (Second Fraction) and sets the corresponding flag (Time Flag) in the GAm message. See Section 6.2.3 for details.

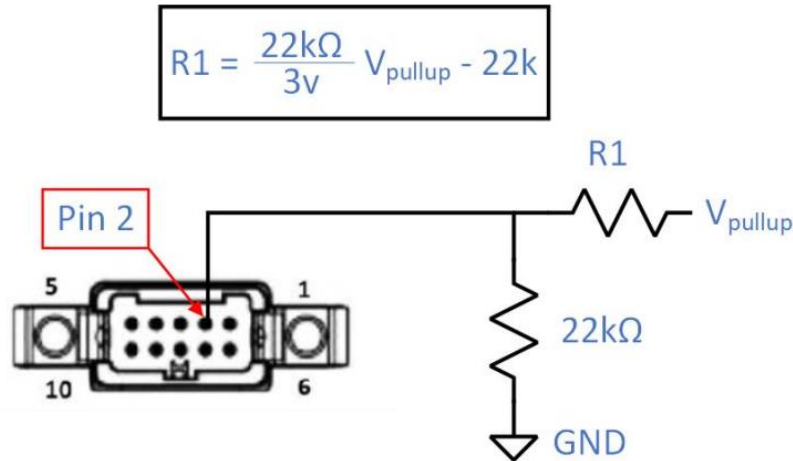


Figure 5.2.

Table 5.2. Electrical specifications.

Power supply KERNEL-100				
	Min	Typical	Max	Units
Supply voltage	+4.0	+5.0	+15.0	VDC
Power	320	365	630	mW
Current	80	73	42	mA
Power Supply KERNEL-108, KERNEL-110, KERNEL-111, KERNEL-120, KERNEL-201, KERNEL-210 and KERNEL-220				
	Min	Typical	Max	Units
Supply voltage	+4.8		+36	VDC
Power	400		600	mW
Current	80		16	mA
Timing				
Start-up time	<200			ms
RS422 Bit-Rate	2M			bits/s
Internal Update Rate	2000			Hz
Max Output Data Rate	2000			Hz
Data latency	The data latency value depends on the KERNEL type. Refer to the device datasheet to find out this value.			-
RS422 interface				
Start Bit	1			bit
Data Length	8			bits
Parity	None			
Stop Bits	1			bit
EXTRIG pin				
Low level Input	0.9			V
High level Input	2.1			V
Trigger Edge	Rising			-
Level duration	1			us
TOV pin				
Low Level Output			0.4	V
High Level Output	2.6			V
Rise and Fall Time			30	ns

Notes:

1. Data latency is the time between the moment the sensors are sampled and the moment the IMU starts transmitting the frame with those sensor data. This time does not include the transmission time.
2. EXTRIG can have any duty cycle, but the duration of any level should not be shorter than specified, and the top of second be should marked with the rising edge of the signal. EXTRIG pin has internal 10k pull-up to 3.3V.
3. TOV pin provides a signal that indicates when the sensors are sampled and when the samples complete transmission. TOV rising edge occurs when the sensors are sampled for the next frame. TOV falling edge occurs after the last bit of the previous frame has finished transmission. Note that since data latency plus transmission time is longer than the time between samples, the falling edge of the TOV signal occurs during transmission of the previous sample. TOV pin has internal 10k pull-up to 3.3V. TOV is not used when EXTRIG is active.

Additionally, the KERNEL IMU can be connected to a host computer with the supplied cable set that is shown in Figure 5.3 (showing mating option 2 as described in Figure 4.2).

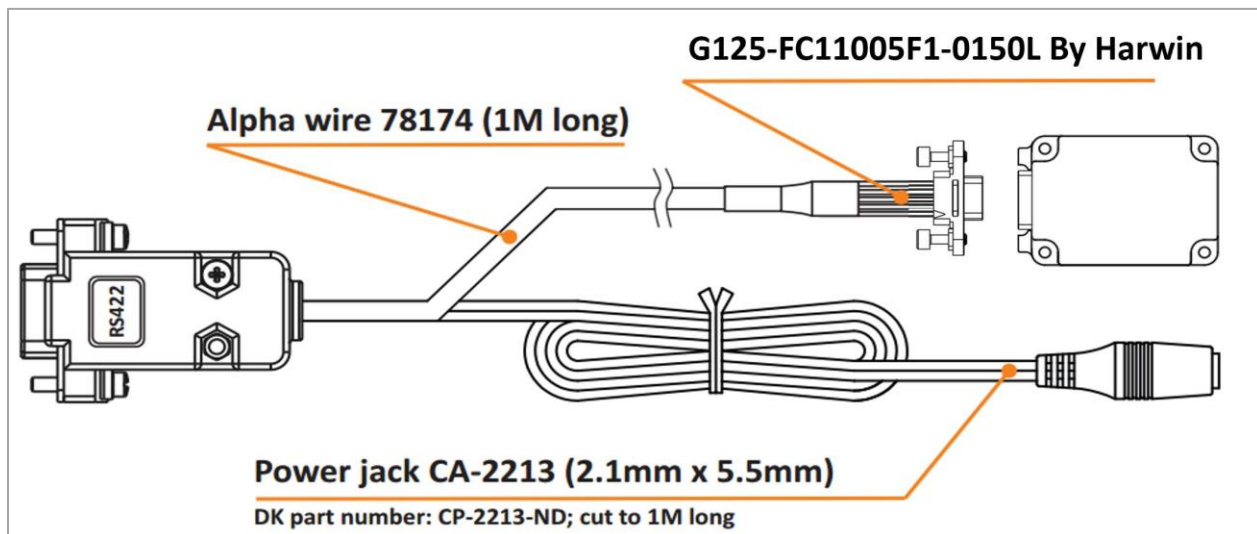


Figure 5.3. RS-422 to mating/power cable for connection of the Inertial Labs KERNEL IMU to host computer.

The KERNEL IMU to PC connection kit is provided with every IMU and includes:

- one interface cable for the IMU connection to a host system equipped with an RS422 interface, with branch wires for DC power source connection
- one RS422-to-USB converter (cable 2) for connection to a PC via USB port (drivers for the RS422-to-USB converter are provided as well);
- one 100-240VAC/4VDC power adapter.

5.1.1. TOV Description

The Inertial Labs KERNEL IMU provides a Time of Validity (TOV) output. See Table 5.1 for a description of the respective pin that is responsible for this output.

The main purpose of the TOV signal is to provide the user with the signal that identifies when the device is ready to transmit data, or when data is being transmitted. The raising edge occurs when data transmission begins, and the falling edge occurs when transmission of a sample ends (see Figure 5.4).

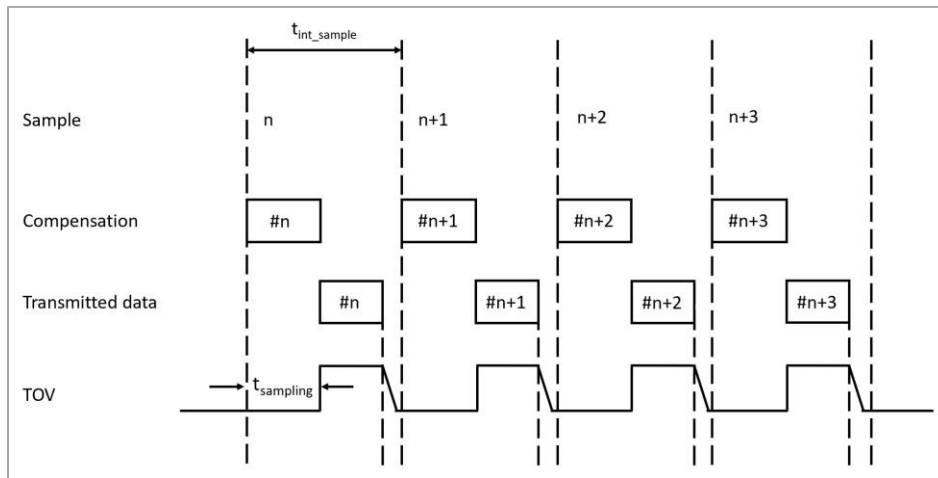


Figure 5.4. Timing of TOV at 1kHz output data rate.

Note: $t_{sampling}$ was tested to be 300 microseconds.

5.2. Ethernet Configuration

By request the device can be equipped with an Ethernet interface. In this case the sensor is supplied inside the bigger enclosure with 24-pin circular connector. Please see below.

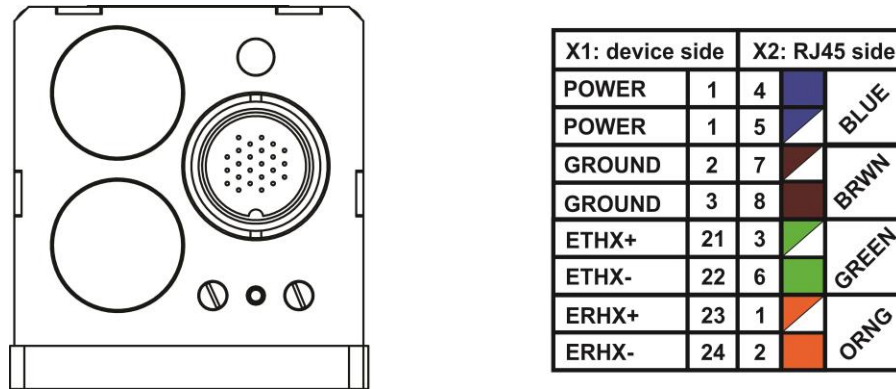


Figure 5.5.

The supplied cable assembly provides Ethernet and power via a single RJ45 connector. For the testing purposes the PoE adapter (so called PoE injector) is also supplied. This cable is shown on the picture below:

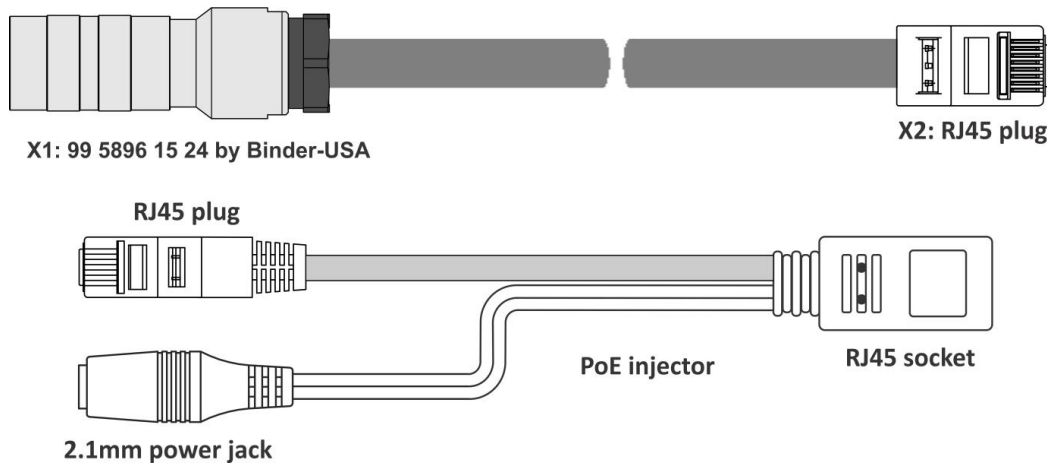
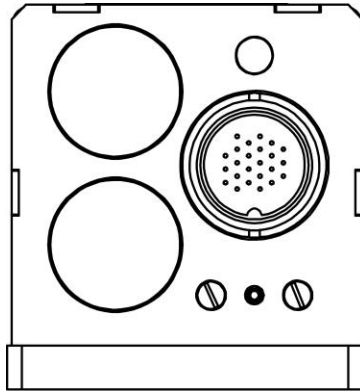


Figure 5.6.

For detailed description about Ethernet interface see Appendix A: “Optional KERNEL Device with Ethernet Interface”.

5.3. Multi-Interface Configuration

By request the device can be equipped with multi-interface configuration. In this case the sensor is supplied inside the bigger enclosure with 24-pin circular connector. Please see below.



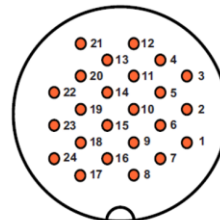
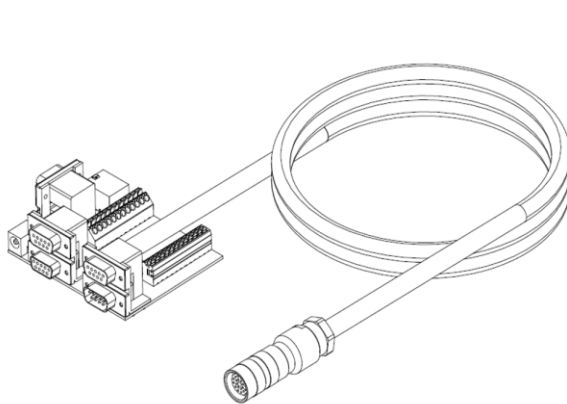
POWER	1
GROUND	2
GROUND	3
TRIGGER	4
TOV	5
GROUND	6
RS422-TX+	11
RS422-TX-	12

RS422-RX-	13
RS422-RX+	14
RS232-RX	17
RS232-TX	18
ETHX+	21
ETHX-	22
ERHX+	23
ERHX-	24

Figure 5.7.

This cable is shown on the picture below:

CAS-01-2M-01-01-V0.5



99 5896 15 24 by Binder-USA
Soldering side view

POWER	1	ORNG	RS422-RX-	13	VIOL
GROUND	2		RS422-RX+	14	
GROUND	3	RED	DNC	15	DK GR
TRIGGER	4		DNC	16	
TOV	5	BRWN	RS232-RX	17	LT GR
GROUND	6		RS232-TX	18	
GROUND	7	DK BL	DNC	19	YELL
GROUND	8		DNC	20	
DNC	9	WHITE	ETHX+	21	PINK
DNC	10		ETHX-	22	
RS422-TX+	11	GRAY	ERHX+	23	LT BL
RS422-TX-	12		ERHX-	24	

Note 1: The cable type is L-com CSTP125-100, total length is 2 meters.

The jacket is removed from the last 3 inches of the open end. Tips of all wires are stripped and tinned.

Note 2: The shield of the cable is grounded on the connector side.

Note 3: TOV - 3.3V TTL time of validity output; EXTRIG - 3.3V TTL external trigger input.

Note 4: The names of the signals are given relative to device, i.e. RS232-RX lines are for data coming to device, while RS232-TX lines are for data coming from the device. Same for RS422.

Figure 5.8.

5.4. Configuration with MIL-STD-1275 Protection (KERNEL-110M)

The KERNEL-110M IMU can be equipped with one of the following connectors, depending on the application (pinout is the same, see Table 5.3):

- Amphenol Aerospace, part # D38999/23YB35PN - 13 PINS (marine application),
- Amphenol Aerospace, part # D38999/24WB35PN - 13 PINS (land application).

Figure 5.9. KERNEL-110M connector pinout (mating side view)

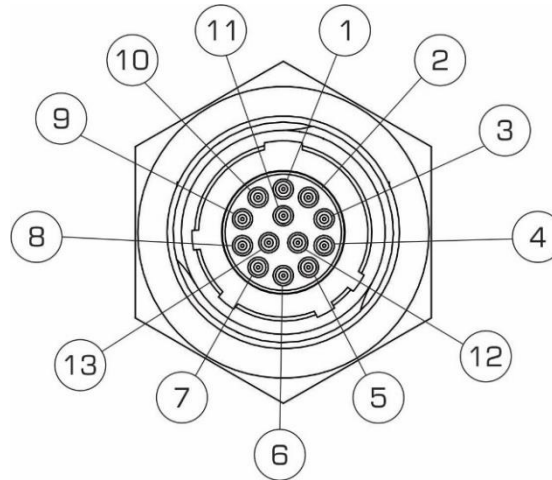


Table 5.3. KERNEL-110M connector pinout

Pin No.	Signal Name	Description	Direction	Source	Destination	Notes
1	EXTRIG	Discrete	Input	Host	IMU	Internally pulled to 4VDC via 100K
2	RS422_RXN	RS422	Input	Host	IMU	
3	1PPS*	Discrete	Input	Host	IMU	Internally pulled to 4VDC via 100K
4	RS422_RXP	RS422	Input	Host	IMU	
5	RS422_TXN	RS422	Output	IMU	Host	
6	RS422_TXP	RS422	Output	IMU	Host	
7	Reserved					
8	Reserved					
9	BOOT	Discrete	Input	Host	IMU	Internally pulled to 4VDC via 10K Pull down for firmware update. Leave floating if not used.
10	Reserved					
11	TOV	Discrete	Output	IMU	Host	Internally pulled down via 10K Output logic low – 0V Output logic high – SUPPLY_IN
12	SUPPLY_RTN	Power	Return	Host	IMU	
13	SUPPLY_IN	Power	Input	Host	IMU	

* 1PPS is optional, confirm availability before ordering

Table 5.4. Electrical specifications.

Power supply KERNEL-110M				
	Min	Typical	Max	Units
Supply voltage	18 to 32			VDC
Power	590 @ 18V	710 @ 28V	810 @ 32V	mW
Electrical standards				
EMC/EMI	MIL-STD-461E			
Protection	MIL-STD-1275			

6. Software Interface

By default, in the factory configuration, commands are transmitted through a serial port according to the protocol RS-422 with baud rate 115200 bits per second. Parameters for the serial port connection are shown in Table 6.1.

Table 6.1. COM-port parameters

COM-port parameters	
Baud rate	115200
Data bits	8
Parity	none
Stop bits	1

Note: Other baud rates than 115200 bps can be set for the IMU, see section 6.3.7 for details.

All commands and messages to/from the Inertial Labs KERNEL IMU have a byte structure as shown in the Table 6.2.

Table 6.2. Byte structure for all commands and messages to / from the KERNEL IMU

Byte number	0	1	2	3	4, 5	6 .. (n+5)	n+6, n+7
Parameter	Header 0	Header 1	Message type	KERNEL data identifier	Message length	Payload	Check sum
Length	1 byte	1 byte	1 byte	1 byte	1 word	n bytes (variable)	1 word
Note	0xAA	0x55		In KERNEL messages	n+6		

Note: From henceforth the following definitions will be used for Table 6.2 and in all other tables:

- **word** = unsigned 2-byte integer;
- **sword** = signed 2-byte integer.

Message type is equal to:

- 0 – for commands;
- 1 – for transferring data.

All the KERNEL IMU outputs are data, therefore they have Message type = 1.

The KERNEL data identifier is used for KERNEL data output only. This byte is expressed as a code, representative of a command that was sent from host system.

Descriptions of all command codes can be seen in “Table B.1. List of the KERNEL commands with exact structure”.

The Message length is the number of bytes in the message not including the header.

The Check sum is the arithmetical sum of bytes 2...(n-1) (all bytes without header). In the check sum LSB (Least Significant Byte) is transmitted first (see Table 6.3).

Table 6.3. Format of the check sum transmitting

byte0	byte1
low byte	high byte

Important note: In all multi-byte data LSB is transmitted first.

6.1. Operational Modes

The KERNEL IMU can operate in the following operating modes:

Idle mode

All sensors and electronics are powered. The IMU microprocessor awaits any command from the host computer to start operation.

Continuous mode

In this mode, the IMU operates in an endless loop, providing the continuous output data according to chosen output data format (see supported data formats in section 6.2).

Firmware update mode

This mode allows for IMU firmware updating. See a separate document provided by Inertial Labs for details on firmware updates.

6.2. Data Output Formats

The following output data formats are available during IMU operation:

- KERNEL Orientation Data (See Section 6.2.1)
- KERNEL GA Data (See Section 6.2.2)
- KERNEL GAm Data (See Section 6.2.3)
- KERNEL Quaternion Data (See Section 6.2.4)
- KERNEL Calibrated HR Data (See Section 6.2.5)
- KERNEL GAA Data (See Section 6.2.6)
- KERNEL GAAM Data (See Section 6.2.7)
- KERNEL User Defined Data (See Section 6.2.8).

6.2.1. KERNEL Orientation Data

This is the default data format. It provides data output from the KERNEL IMU in the form of:

- 3 orientation angles (heading, pitch, and roll);
- calibrated outputs of the 6 sensors (gyros and accelerometers) that give information about current angular rates, and linear accelerations of the IMU.

Structure of the IMU data blocks for the “KERNEL Orientation Data” format corresponds to Table 6.2 with payload as shown in Table 6.4.

Table 6.4. The message payload at KERNEL Orientation Data format

Byte number	0 – 1	2 – 3	4 – 5	6 – 11	12 – 17	18 – 23
Parameter	Heading	Pitch	Roll	GyroX, GyroY, GyroZ	AccX, AccY, AccZ	MagX, MagY, MagZ
Length	2 byte word	2 byte sword	2 byte sword	3×2 byte sword	3×2 byte sword	3×2 byte sword
Note	Orientation angles, deg*100			Angular rates, deg/s*KG	Accelerations, g*KA	0 for the IMU (not measured)

Table 6.4 (continued)

Byte number	24–27	28 – 29	30 – 31	32 – 33
Parameter	Reserved	USW	Reserved	Temper
Length	4 byte word	2 byte word	2 byte word	2 byte sword
Note		See Note 5.		Temperature, °C*10

Notes:

1. $g = 9.8106 \text{ m/s}^2$.
2. Values of KG, KA are scale factors for KERNEL Orientation Data are dependent on gyro and accelerometer range (see Table 6.5).
3. Angular rates and linear accelerations are in a carrier object axes (X is lateral axis, Y is longitudinal axis, Z is vertical axis). The IMU orientation relative to the carrier object axes is set by alignment angles (see Appendix C. Variants of the IMU Mounting Relative to the Object Axes).
4. Gyros and accelerometers data can be presented in form of instant, average or incremental values.
5. USW is a unit status word (see Section 6.4 The Unit Status Word definition”).
6. Temper is averaged temperature in 3 gyros.
7. LSB is transmitted first.

Table 6.5. Values of KG and KA factors

Gyro range, deg/sec	450	950	2000
KG	50	20	10
Accelerometer range, g	8	15	40
KA	4000	2000	500

6.2.2. KERNEL GA Data

The KERNEL GA Data format provides the IMU calibrated output of gyros and accelerometers with high resolution and contains information about current angular rates and linear accelerations of the KERNEL IMU.

Structure of the IMU data blocks using the “KERNEL GA Data” format corresponds to Table 6.2 with a payload as shown in Table 6.6.

Table 6.6. The message payload at KERNEL GA Data format

Byte number	0 – 11	12 – 23	24 – 25	26 – 27	28 – 29	30 – 31
Parameter	GyroX, Gyro Y, GyroZ	AccX, AccY, AccZ	Counter	USW	Reserved	Temper
Length	3×4 byte integer	3×4 byte integer	2 byte word	2 byte word	2 byte word	2 byte sword
Note	Angular rates, deg/s*1.0e5	Accelerations, g*1.0e6	See Note 3	See Note 4		Temperature, °C*10

Notes:

1. $g = 9.8106 \text{ m/s}^2$.
2. Angular rates and linear accelerations are in the carrier object axes (X is lateral axis, Y is longitudinal axis, Z is vertical axis). The IMU orientation relative to the carrier object axes is set by alignment angles (see “Appendix C. Variants of the IMU Mounting Relative to the Object Axes”).
3. The counter increments with the frequency of 2 kHz (frequency of the internal computations of the device).
4. USW is the unit status word (see section 6.4 for details).
5. LSB is transmitted first.

6.2.3. KERNEL GAm Data

The KERNEL GAm Data format provides the KERNEL IMU calibrated output of gyros and accelerometers with high resolution and contains information about current angular rates and linear accelerations of the IMU.

Structure of the IMU data blocks using the “KERNEL GAm Data” data format corresponds to Table 6.2 with a payload shown in Table 6.7.

Table 6.7. Message payload structure for the KERNEL “KERNEL GAm Data” format

Byte number	0 – 11	12 – 23	24 – 25	26 – 29	30 – 31
Parameter	GyroX, GyroY, GyroZ	AccX, AccY, AccZ	Time Flag	Second fraction	Temperature
Length	3×4-byte integer	3×4-byte integer	2-byte unsigned integer	4-byte unsigned integer	2-byte signed integer
Note	Angular rates, deg/s*1.0e5	Accelerations, g*1.0e6	See Note 2	Second fraction, nsec See Note 3	Temperature, °C*10

Notes:

1. $g = 9.8106 \text{ m/s}^2$.

2. The time flags are as follows:
 - 0x0001 – 1PPS was received between the last frame and this frame
 - 0x0008 – 1PPS was lost – no 1PPS for more than 1 second.
3. Second fraction:
 - Before the first 1PPS is received, the second fraction values are aligned to IMU power-on.
 - Once 1PPS is received, all future second fraction values are aligned to that 1PPS.
 - Once 1PPS is lost, drift of second fraction is possible.
 - Once 1PPS is received again, the second fraction is immediately re-aligned to it.
4. LSB is transmitted first.

6.2.4. KERNEL Quaternion Data

The “KERNEL Quaternion Data” format provides data in the following format:

- Euler Orientation Angles
- Quaternion of Orientation.

Note: See “Appendix D. Forms of the IMU Orientation Presentation” for the correct relationship between orientation angles and quaternion.

Structure of the IMU data blocks at the “KERNEL Quaternion Data” data format corresponds to the Table 6.2 with payload shown in Table 6.8.

Table 6.8. The message payload for the KERNEL Quaternion Data format

Byte number	0 – 1	2 – 3	4 – 5	6 – 13	14 – 27	28 – 29	30 – 31	32 – 33
Parameter	Heading	Pitch	Roll	q0, q1, q2, q3	Reserved	USW	Reserved	Temper
Length	2 byte word	2 byte sword	2 byte sword	4x2 byte sword	14 bytes	2 byte word	2 byte word	2 byte sword
Note	Orientation angles, deg*100			Quaternion of orientation *10000		See Note 2.		Temperature, °C*10

Notes:

1. $g = 9.8106 \text{ m/s}^2$.
2. USW is a unit status word (see Section 6.4 The Unit Status Word definition”).
3. Temper is averaged temperature in 3 gyros.
4. LSB is transmitted first.

6.2.5. KERNEL Calibrated HR Data

The “KERNEL Calibrated HR Data” format provides data from the KERNEL in the form of:

- 3 orientation angles (relative heading, pitch and roll) in high resolution
- calibrated outputs of the 6 sensors (gyros, accelerometers) that give information about current angular rates, and linear accelerations of the KERNEL in high resolution.
- A counter
- USW and Temperature

Structure of the KERNEL data blocks using the “KERNEL Calibrated HR Data” format corresponds to Table 6.2 with payload as shown in Table 6.9.

Table 6.9. The message payload for the KERNEL Calibrated HR Data format

Byte number	0 – 3	4 – 7	8 – 11	12 – 23	24 – 35	36 – 41
Parameter	Heading	Pitch	Roll	GyroX, GyroY, GyroZ	AccX, AccY, AccZ	MagX, MagY, MagZ
Length	4 byte integer	4 byte integer	4 byte integer	3×4 byte integer	3×4 byte integer	3×2 byte sword
Note	Orientation angles, deg*1000			Angular rates, deg/s*1e5	Accelerations, g*1e6	0 for the IMU (not measured)

Table 6.9 (continued)

Byte number	42 – 43	44 – 45	46 – 47	48 – 49	50 – 51
Parameter	Counter	Reserved	USW	Reserved	Temper
Length	2 byte sword	2 byte sword	2 byte word	2 byte word	2 byte sword
Note	See Note 8.		See Note 4.		Temperature, °C*10

Notes:

1. Angular rates and linear accelerations are in the carrier object axes (X is lateral axis, Y is longitudinal axis, Z is vertical axis). The IMU orientation relative to the carrier object axes is set by alignment angles (see Appendix C. Variants of the IMU Mounting Relative to the Object Axes).
2. $g = 9.8106 \text{ m/s}^2$.
3. USW is unit status word (see Section 6.4 The Unit Status Word definition”).
4. Temper is averaged temperature in 3 gyros.
5. LSB is transmitted first.
6. The counter increments with the frequency of 2kHz (frequency of internal computations of the KERNEL).

6.2.6. KERNEL GAA Data

Note: This output data format is ONLY available for the KERNEL-120 and KERNEL-220 models.

The KERNEL GAA Data format is based on the KERNEL GA Data format. The only difference is that to the message structure added data from two additional accelerometers.

The structure of the transmitted messages corresponds to Table 6.2 with a payload shown in Table 6.10.

Table 6.10. The message payload at KERNEL GAA Data format

Byte number	0 – 11	12 – 23	24 – 35	36 – 37	38 – 39	40 – 41	42 – 43
Parameter	GyroX, Gyro Y, GyroZ	Acc1X, Acc1Y, Acc1Z	Acc2X, Acc2Y, Acc2Z	Reserved	USW	Reserved	Temper
Length	3×4 byte integer	3×4 byte integer	3×4 byte integer	2 byte word	2 byte word	2 byte word	2 byte sword
Note	Angular rates, deg/s*1.0e5	Accelerations, g*1.0e6	Accelerations, g*1.0e6		See Note 3		Temperature, °C*10

Notes:

1. $g = 9.8106 \text{ m/s}^2$.
2. Angular rates and linear accelerations are in the carrier object axes (X is lateral axis, Y is longitudinal axis, Z is vertical axis). The KERNEL orientation relative to the carrier object axes is set by alignment angles (see “Appendix C. Variants of the IMU Mounting Relative to the Object Axes”).
3. USW is the unit status word (see Section 6.4 The Unit Status Word definition”).
4. LSB is transmitted first.

6.2.7. KERNEL GAAM Data

Note: This output data format is ONLY available for the KERNEL-120 and KERNEL-220 models.

The KERNEL GAAM Data format is based on the KERNEL GAM Data format. The only difference is that to the message structure added data from two additional accelerometers.

The structure of the transmitted messages corresponds to Table 6.2 with a payload shown in Table 6.11.

Table 6.11. Message payload structure for the KERNEL “KERNEL GAAM Data” format

Byte number	0 – 11	12 – 23	24 – 35	36 – 37	38 – 41	42 – 43
Parameter	GyroX, GyroY, GyroZ	Acc1X, Acc1Y, Acc1Z	Acc2X, Acc2Y, Acc2Z	Time Flag	Second fraction	Temperature
Length	3×4-byte integer	3×4-byte integer	3×4-byte integer	2-byte unsigned integer	4-byte unsigned integer	2-byte signed integer
Note	Angular rates, deg/s*1.0e5	Accelerations, g*1.0e6	Accelerations, g*1.0e6	See Note 2	Second fraction, nsec See Note 3	Temperature, °C*10

Notes:

1. $g = 9.8106 \text{ m/s}^2$.
2. The time flags are as follows:
 - 0x0001 – 1PPS was received between the last frame and this frame
 - 0x0008 – 1PPS was lost – no 1PPS for more than 1 second.
3. Second fraction:
 - Before the first 1PPS is received, the second fraction values are aligned to IMU power-on.
 - Once 1PPS is received, all future second fraction values are aligned to that 1PPS.
 - Once 1PPS is lost, drift of second fraction is possible.
 - Once 1PPS is received again, the second fraction is immediately re-aligned to it.
4. LSB is transmitted first.

6.2.8. User Defined Data

Optionally, a user can configure their own desired output data formats using the “User Defined Data” output data format. The **User_Def_Data** command (see Appendix B “Full List of the IMU Commands” for command details) can be used to request this data format. The available data formats and descriptions are seen below.

Structure of the IMU data blocks at the “User Defined Data” output format corresponds to Table 6.2, with payload shown in Table 6.12.

Table 6.12. Payload of the “User Defined Data”

Field	Offset in payload, bytes	Size, bytes	Value
Pckg Number	0	1	Number of data packages present in the payload (N)
Data List	1	N	List of data types, one per byte (refer to Table 6.13 for values)
Data #1	N+1	Variable, depends on data type (refer to Table 6.14 for sizes)	Data according to structure shown in Table 6.14
Data #2	Variable, depends on Data #1 size	Variable, depends on data type (refer to Table 6.14 for sizes)	Data according to structure shown in Table 6.14
...	...	...	...
Data #N	Variable, depends on preceding data	Variable, depends on data type (refer to Table 6.14 for sizes)	Table 6.14

Table 6.13. Available data for configuration of the “User Defined Data” format.

Data group	Data type	Description
Time data	0x02	IMU time
	0x05	Second fraction (applicable to all KERNEL IMUs except KERNEL-100)
Orientation data	0x07	Orientation angles (Euler)
	0x08	Orientation angles HR (Euler)
	0x09	Quaternion of orientation
Sensors data	0x20	Gyro data
	0x21	Gyro data HR
	0x22	Accelerometer data
	0x23	Accelerometer data HR
	0x2D	Bias-compensated gyro data
	0x2E	Gyro bias
	0x2F	Aux accelerometer (only for KERNEL-120 and KERNEL-220)
	0x91	Filtered low-G accels HR
	0x92	Filtered high-G accels HR
IMU service data	0x50	Supply voltage
	0x52	Temperature

	0x53	Unit status word (USW)
	0x5B	USW Extended

Table 6.14. Detailed description of “User Defined Data” structure.

Data type	Description	Size, bytes	Structure	Note
0x02	IMU time	8	unsigned integer	Time of IMU solution, seconds*1.0e9 from time of power is applied to device.
0x05	Second fraction (applicable to all KERNEL IMUs except KERNEL-100)	4	unsigned integer	The fraction of the second in nsecs. See more information in note 7 below.
0x07	Orientation angles	6	word	Heading, deg*100
			sword	Pitch, deg*100
			sword	Roll, deg*100
0x08	Orientation angles HR	12	unsigned integer	Heading, deg*1000
			integer	Pitch, deg*1000
			integer	Roll, deg*1000
0x09	Quaternion of orientation	8	sword	q0*10000
			sword	q1*10000
			sword	q2*10000
			sword	q3*10000
0x20	Gyro data	6	sword	Gyro X, deg/s*KG (for instant and average data form) or deg*KG (for incremental data form). See Note 2 and Note 3.
			sword	Gyro Y, deg/s*KG (for instant and average data form) or deg*KG (for incremental data form). See Note 2 and Note 3.
			sword	Gyro Z, deg/s*KG (for instant and average data form) or deg*KG (for incremental data form). See Note 2 and Note 3.
0x21	Gyro data HR	12	integer	Gyro X, deg/s*1.0e5 (for instant and average data form) or deg*1.0e5 (for incremental data form). See Note 2.
			integer	Gyro Y, deg/s*1.0e5 (for instant and average data form) or deg*1.0e5 (for incremental data form). See Note 2.
			integer	Gyro Z, deg/s*1.0e5 (for instant and average data form) or deg*1.0e5 (for incremental data form). See Note 2.
0x22	Accelerometer data	6	sword	Accelerometer X, g*KA (for instant and average data form) or g*s*KA (for incremental data form). See Note 2 and Note 3.
			sword	Accelerometer Y, g*KA (for instant and average data form) or g*s*KA (for incremental data form). See Note 2 and Note 3.
			Sword	Accelerometer Z, g*KA (for instant and average data form) or g*s*KA (for incremental data form). See Note 2 and Note 3.

0x23	Accelerometer data HR	12	integer	Accelerometer X, $g \cdot 1.0e6$ (for instant and average data form) or $g \cdot s \cdot 1.0e6$ (for incremental data form). See Note 2.
			integer	Accelerometer Y, $g \cdot 1.0e6$ (for instant and average data form) or $g \cdot s \cdot 1.0e6$ (for incremental data form). See Note 2.
			integer	Accelerometer Z, $g \cdot 1.0e6$ (for instant and average data form) or $g \cdot s \cdot 1.0e6$ (for incremental data form). See Note 2.
0x2D	Bias-compensated gyro data	12	integer	Gyro X, $deg/s \cdot 1.0e5$
			integer	Gyro Y, $deg/s \cdot 1.0e5$
			integer	Gyro Z, $deg/s \cdot 1.0e5$
0x2E	Gyro Bias	12	integer	Gyro X bias, $deg/s \cdot 1.0e5$
			integer	Gyro Y bias, $deg/s \cdot 1.0e5$
			Integer	Gyro Z bias, $deg/s \cdot 1.0e5$
0x2F	Aux accelerometer (only for KERNEL-120 and KERNEL-220)	12	integer	Aux accelerometer X, $g \cdot 1.0e6$
			integer	Aux accelerometer Y, $g \cdot 1.0e6$
			integer	Aux accelerometer Z, $g \cdot 1.0e6$
0x91	Filtered low-G accels HR	12	integer	Filtered low-G accelerometer X, $g \cdot 1.0e6$
			integer	Filtered low-G accelerometer Y, $g \cdot 1.0e6$
			integer	Filtered low-G accelerometer Z, $g \cdot 1.0e6$
0x92	Filtered high-G accels HR	12	integer	Filtered high-G accelerometer X, $g \cdot 1.0e6$
			integer	Filtered high-G accelerometer Y, $g \cdot 1.0e6$
			integer	Filtered high-G accelerometer Z, $g \cdot 1.0e6$
0x50	Supply voltage	2	word	VDC*100
0x52	Temperature	2	sword	$^{\circ}C \cdot 10$
0x53	Unit status word	2	word	USW (see Note 1)
0x5B	Unit status word extended	2	word	USW Extended (see Note 1)

Notes:

1. Unit status word (USW) is highly recommended for choice to control IMU unit state (see Section “6.4. The Unit Status Word definition”).
2. Data from gyros and accelerometers can be presented in the form of instant, average, or incremental values (see description of each form in section 6.3.6).
3. Values of KG, KA are scale factors depending on gyro and accelerometer range (see Table 6.5).
4. $g = 9.8106 \text{ m/s}^2$.
5. Angular rates and linear accelerations are in the carrier object axes (X = lateral axis, Y = longitudinal axis, Z = vertical axis). The IMU orientation relative to the carrier object axes is set by alignment angles (see “Appendix C. Variants of the IMU Mounting Relative to the Object Axes”).
6. Temperature is averaged temperature in 3 gyros.
7. Second fraction:
 - Before the first 1PPS is received, the second fraction values are aligned to IMU power-on.
 - Once 1PPS is received, all future second fraction values are aligned to that 1PPS.

- Once 1PPS is lost, drift of second fraction is possible.
 - Once 1PPS is received again, the second fraction is immediately re-aligned to it.
8. LSB is transmitted first.

Use the **User_Def_Data_config** command (see Section 6.3.5. User Defined Data Commands”) to choose data types for output. Because the package length of the “User Defined Data” is variable and depends on user choice of data, then the maximum data rate for the KERNEL output is also variable. It can be estimated using formula 6.1 in Section 6.5. Limitation of the IMU Maximum Output Data Rate.

The KERNEL unit controls set data rate. If it exceeds the maximum value of the currently set COM-port baud rate and length of chosen data, then the KERNEL will automatically decrease the set data rate to the allowed value and will inform the user about this change in response to the **User_Def_Data_config** command (see Section 6.3.5 User Defined Data Commands).

Send the **Get_User_Def_Data_struct** command to check the “User Defined Data” structure set by the previous user (see Section 6.3.5 User Defined Data Commands).

6.3. Control of the IMU

The following commands are used to control the KERNEL IMU:

- KERNEL_Orientation
- KERNEL_GAData
- KERNEL_GAmData
- KERNEL_GAADData
- KERNEL_GAAmData
- KERNEL_QuatData
- KERNEL_CalibHR
- User_Def_Data
- Stop
- GetDevInfo
- GetBIT
- SaveFlash
- ReadRAM.

All these commands have a byte structure as shown in Table 6.2. The payloads for these commands have lengths of 1 byte and contain a code identifying the command. See: “Table B.1. List of the KERNEL commands with exact structure” for the exact structure of these commands.

6.3.1. KERNEL_Data Commands

The Data Commands are used to start the Inertial Labs KERNEL IMU in the “Continuous” operating mode with the appropriate variant of an output data format as shown in Table 6.15.

Table 6.15. KERNEL control commands and appropriate output data format

Command	Code	Output data format	
		Name	Structure
KERNEL_Orientation	0x33	KERNEL Orientation Data	Inertial Labs binary structure according to Table 6.2
KERNEL_GAData	0x8F	KERNEL GA Data	
KERNEL_GAMData	0x9B	KERNEL GAM Data	
KERNEL_GAData	0xA5	KERNEL GA Data	
KERNEL_GAMData	0xA6	KERNEL GAM Data	
KERNEL_QuatData	0x82	KERNEL Quaternion Data	
KERNEL_CalibHR	0x81	KERNEL Calibrated High Resolution Data	
User_Def_Data	0x95	User Defined Data	

All these commands have a byte structure as shown in Table 6.2. The payload for all commands has a length of 1 byte and contain a code of the command listed in Table 6.15.

The KERNEL IMU acknowledges receiving any of these commands by responding back immediately. This response is the calculated check sum of the message received (without its header and check sum) and it is returned to the host system for user verification. The byte structure of such message is shown in Table 6.2 where the payload is a calculated check sum (1 word). This check sum should be equal to the check sum that was sent to the IMU.

After acknowledgement, for output data formats that include orientation or quaternion angles, the KERNEL IMU starts the process of running an initial alignment algorithm (only applies for data formats that output orientation angles) which by default takes 2 seconds (should be increased, if possible, maximum needed is 30 seconds). The initial alignment process includes the KERNEL IMU estimating gyro biases and fusing inertial data to calculate orientation, therefore don't move the IMU during its initial alignment. If this requirement is disregarded, then accuracy of orientation solution will be decreased.

Note: If the device is started while the carrier object is in motion or for some other reason it is impossible to achieve absolutely stationary standing at the start, you should set the initial alignment time to 0 seconds (see Section 6.3.7, or change in the GUI, See IMU GUI User Manual) to skip initial alignment. As a result, the KERNEL IMU dynamic accuracy may be decreased.

After completing of the initial alignment (only used for data formats that include orientation or quaternion angles), the IMU goes into the “Continuous” operating mode.

In the “Continuous” operating mode, set by any of commands `KERNEL_Data` commands, the program in the IMU microprocessor operates in an endless loop, providing the process of data reading from ADC and calculating orientation angles.

6.3.2. Stop Command

Upon receiving the Stop command (code 0xFE in the “Payload” field) the KERNEL IMU stops working in the operating mode and goes into the idle mode. The KERNEL IMU is then ready to receive any additional command from a host computer.

6.3.3. GetDevInfo Command

The GetDevInfo command (code 0x12 in the “Payload” field) is used to get detailed information about the KERNEL IMU.

As a response, the IMU sends out a message with structure according to Table 6.2 and payload shown in Table 6.16.

Table 6.16. Payload of the KERNEL response to the GetDevInfo command

Byte	Parameter	Format	Length	Note
0-7	ID_sn	char	8	IMU device (KERNEL) s/n
8-47	ID_fw	char	40	IMU (KERNEL) firmware version
48	Reserved	byte	1	
49	IMU_type	byte	1	IMU type
50-57	IMU_sn	char	8	IMU device (KERNEL) s/n (duplicated)
58-97	IMU_fw	char	40	IMU (KERNEL) firmware version (duplicated)
98-165	Reserved	byte	68	

6.3.4. GetBIT Command

The GetBIT command (code 0x1A in the “Payload” field) is used to identify the IMU current health status. In response, the IMU sends out a message with data according to the Table 6.17.

Table 6.17. Payload of the KERNEL response to the GetBIT command

Byte	Bit	Parameter	Description
0 (Low byte)	0	Accel BIT Result	0 – Ok
	1		2 – Ok Anything Else – Report Issue to Inertial Labs for Analysis
	2	Gyro BIT Result	0 – Ok Anything Else – Report Issue to Inertial Labs for Analysis
	3		
	4		
	5		
	6		
	7		
1	8	Reserved	
	9		
	10	Reserved	
	11		
	12	Flash Check	0 – Ok
	13		Anything Else – Report Issue to Inertial Labs for Analysis
	14	Reserved	
	15		
2	16-23	Reserved	
3 (High byte)	24-31	Reserved	

6.3.5. User Defined Data Commands

To configure the data set that the user would like to configure the device to output in the “User Defined Data” output, the **User_Def_Data_config** command (code 0x96 in the “Payload” field) should be sent to the device, followed by the message with the block of parameters listed in Table 6.18. This message has the byte structure shown in Table 6.2 and should be sent without delay after sending the User_Def_Data_config command.

Table 6.18. Payload for the message sent after sending the User_Def_Data_config command and response after sending Get_User_Def_Data command.

Byte	Parameter	Format	Length	Note
0	Pckg Number	byte	1	Number of data packages for output (N)
1 to N	Data List	byte	N	List of data types, one per byte (refer to Table 6.13 for values)

The KERNEL calculates the check sum of the received message with parameters and returns it for user validation. Additionally, the KERNEL checks validity of the list of data types. Byte structure of the KERNEL response corresponds to Table 6.2, where the payload is shown in Table 6.19.

Table 6.19. Payload for KERNEL response after sending the User_Def_Data_Config command.

Byte number	0 – 1	2	3 – 4	5 - 6	7
Parameter	Check sum of received message	Data list error	Data types correctness	Maximum data rate	Reserved
Length	1 word	1 byte	1 word	1 byte	1 byte

Notes:

1. Calculated check sum in bytes 0 – 1 should be equal to the check sum of the message with payload Table 6.18 that was sent to the KERNEL. Otherwise, the User_Def_Data_config command is ignored.
2. “Data list error” can have values listed in Table 6.20.
3. If any bit from 0-2 in Table 6.20 is returned as a 1, then the User_Def_Data_config command is ignored. In such a case, the check sum of the message after this command is not calculated and bytes #0 and #1 in the KERNEL response Table 6.19 are zero.
4. The User_Def_Data_config command is still accepted if bit #3 is either a 1 or 0 (from Table 6.20). The same applies to other data formats; after the KERNEL starts, it checks compliance of data package size with set data rate and COM-port baud rate. In the event of an incompliance, the KERNEL data rate will be corrected to an allowed (lower) value.
5. “Data types correctness” indicates correctness of each ordered data type for up to 16 first types. If i-th data type is correct, then i-th bit of the “Data list correctness” is

zero. Otherwise, the i-th bit is set to 1. If “Data list correctness” is not zero, the User_Def_Data_config command is ignored.

6. “Maximum data rate” shows the maximum data rate (Hz) for user defined data at current baud rate.

Table 6.20. Codes for indicating data error.

Bit	Value and description
0	1 – incorrect structure of the message following the User_Def_Data_config command, with payload listed in Table 6.18 0 – no errors in the message structure
1	1 – incorrect data in the payload (Table 6.18) of the message following the User_Def_Data_config command 0 – no errors in the payload
2	1 – wrong data types are ordered 0 – no errors in data types. Wrong data types are shown in the “Data list correctness” field of the KERNEL response (Table 6.19)
3	1 – ordered list of data is too large for current output data rate and COM-port baud rate; 0 – size of KERNEL output data meets current data rate and COM-port baud rate

Additionally, the **Get_User_Def_Data_struct** command (code 0x97 in the “Payload” field) can be sent from the host computer to check the “User Defined Data” structure previously set. As a response to this command, the KERNEL sends out a message with information shown in Table 6.18. This message has the byte structure as shown in Table 6.2.

6.3.6. IMU Output Variant

IMU can output data in the following types:

- **Instant sensors data** - outputs the exact, instant value as read by the accelerometers and gyroscopes for all three axes.
- **Average sensors data** - outputs the averaged values of the accelerometers and gyroscopes for all three axes. Computed averages are calculated based on internal algorithm rate (2kHz). I.e. – if output data rate of 2kHz is selected for device, then these values will be “Instant”; however, if the user outputs data at 500Hz then the device will average four values (per sensor, per axis) together and output a singular averaged value.
- **Incremental sensors data** - outputs the delta (change in) each angular rate, or acceleration based on the previously output value. In fact, gyros and accelerometer data are presented here as angle increment (Delta Theta, deg) and velocity increment (Delta Velocity, g*sec) between two data messages divided by the time step of data output.

The IMU data variant can be set using SaveFlash command. To do this, you need to modify the Output_variant parameter (see Table 6.23). Alternatively, you can use the IMU GUI software to configure IMU (refer to the User Manual, section 4.5.4 for details).

6.3.7. SaveFlash Command

The SaveFlash command allows the user to change the value of the individual parameter directly in the IMU flash memory.

The list of available parameters for change are shown in Table 6.23. The SaveFlash command structure can be seen in Table 6.21.

Important note: In all multi-byte data LSB is first.

Table 6.21. The SaveFlash command structure

Field #	1	2	3	4
Bytes Offset	0 – 3	4 – 5	6 – (n-1)	N – (n+1)
Data type	integer	word		word
Size	4	2	Variable	2
Description	Header	Command length (n)	Payload	Check sum
Value	0xAA 0x55 0x00 0x00	Variable	Variable (see Table 6.22)	Variable

Field 1 (Header)

The header has a permanent value in the SaveFlash command. To ensure the command sends, do not change the header.

Field 2 (Command length)

“Command length (n)” = (payload length + 6) or (whole packet length – 2).

Field 3 (Payload)

Table 6.22. The SaveFlash command Payload structure

Field #	3.1	3.2	3.3	3.4
Bytes Offset	0 – 1	2 – 3	4 – 5	6 – ...
Data type	word	word	word	
Size	2	2	2	Variable
Description	Command ID	Changing parameter ID	Changing parameter length	Changing parameter value
Value	0xB2 0xFF	Variable (see Table 6.23)	Variable (see Table 6.23)	Variable (see Table 6.23)

Table 6.23. Available parameters for change

Parameter name	Parameter ID (HEX, LSB)	Parameter length (HEX, LSB)	Parameter value	Description	
1. Devices options					
1.1. IMU					
Data_rate	0x12 0x00	0x02 0x00	Word (2 byte)	Output data rate in Hertz (Hz). Minimal value of the parameter is 1 Hz	
Initial_alignment_time	0x14 0x00	0x02 0x00	Word (2 byte)	The initial alignment time in seconds. Default value is set to 0 seconds.	
Alignment_angles	0x3A 0x02	0x0C 0x00	3*float (12 byte)	Angles of IMU mounting on the carrier object in degrees. First heading (yaw), then pitch and then roll (see Appendix C). The default value of each angle is set to 0 degrees.	
Baud_rate	0xB2 0x03	0x01 0x00	byte	Baud rate	
				Value	
				115200 bps	
				4800 bps	
				9600 bps	
				14400 bps	
				19200 bps	
				38400 bps	
				57600 bps	
				115200 bps	
				230400 bps	
				460800 bps	
				921600 bps	
				2000000 bps	
				375000 bps	
				1843200 bps (for specific purposes only, see note 1 below)	
				3686400 bps (for specific purposes only, see note 1 below)	
1000000 bps					
4000000 bps (for specific purposes only, see note 1 below)					
Notes: 1. The 1843200, 3686400, and 4000000 baud rate values were designed for specific purposes only, and a particular adaptor for data/commands transmission is required. Do not apply such values					

				to the IMU without Inertial Labs agreement. 2. If the default baud rate for the IMU COM port is changing, then the same baud rate must also be set for the COM port of the host computer to keep data/commands transmission. 3. The list of supported baud rates may vary from one device to another device. The supported ones depend on the user's needs. In case of any questions related to the baud rate change, please contact the Inertial Labs support team.						
Average_output_data	0xE1 0x03	0x01 0x00	byte	The output of averaged data. Can be used if the output data rate is set to a value less than the maximum value (2000 Hz). <table><tr><th>Type of data</th><th>Value</th></tr><tr><td>Instant (default)</td><td>0x00</td></tr><tr><td>Averaged</td><td>0x01</td></tr></table>	Type of data	Value	Instant (default)	0x00	Averaged	0x01
Type of data	Value									
Instant (default)	0x00									
Averaged	0x01									
Auto_Start	0xA9 0x03	0x01 0x00	byte	The auto start mode allows the unit to start operation and data output after power is applied without any additional commands from the host computer. It is possible to choose the desirable output data format for auto start (see section 6.2). <table><tr><th>State</th><th>Value</th></tr><tr><td>Disabled (default)</td><td>0x00</td></tr><tr><td>Output data format</td><td>0xXX</td></tr></table> 0xXX – code of output data format (see Table 6.15).	State	Value	Disabled (default)	0x00	Output data format	0xXX
State	Value									
Disabled (default)	0x00									
Output data format	0xXX									
1.2. Triggers										
Trigger_control	0xFA 0x05	0x01 0x00	byte	This parameter allows the external trigger input for data synchronization: <table><tr><th>State</th><th>Value</th></tr><tr><td>Disabled (default)</td><td>0x00</td></tr><tr><td>Enabled</td><td>0x01</td></tr></table>	State	Value	Disabled (default)	0x00	Enabled	0x01
State	Value									
Disabled (default)	0x00									
Enabled	0x01									
Polarity	0xFB 0x05	0x01 0x00	byte	Specifies the polarity of the external trigger pulse. <table><tr><th>State</th><th>Value</th></tr><tr><td>Negative (default)</td><td>0x00</td></tr><tr><td>Positive</td><td>0x01</td></tr></table>	State	Value	Negative (default)	0x00	Positive	0x01
State	Value									
Negative (default)	0x00									
Positive	0x01									

Hold-off_time	0xFE 0x05	0x02 0x00	word (2 byte)	A time period, in milliseconds, during which subsequent pulses after an initial pulse are ignored.										
TOV_control	0x99 0x04	0x01 0x00	byte	The TOV signal identifies when data transmission is ready, or when data is being transmitted (see section 5.1.1). In some specific applications, the TOV pin can provides Built-in Test (BIT). Output signal logic of BIT is as follows: 0 - everything is good, 1 - any error flags in USW More information about USW can be seen in section 6.4. <table><tr><th>State</th><th>Value</th></tr><tr><td>TOV (default)</td><td>0x00</td></tr><tr><td>BIT</td><td>0x01</td></tr></table>	State	Value	TOV (default)	0x00	BIT	0x01				
State	Value													
TOV (default)	0x00													
BIT	0x01													
2. Correction options														
2.1. IMU tilt algorithm														
Motion_type	0x12 0x04	0x01 0x00	byte	Parameter for changing specific settings used for correction data in the device's Kalman Filter. For questions on these parameters, please contact the Inertial Labs Support Team.										
Gyro_noise_RMS	0xEE 0x03	0x04 0x00	float											
Gyro_scale_factor error	0xF2 0x03	0x04 0x00	float											
Gyro_bias_drift	0xF6 0x03	0x04 0x00	float											
Gyro_bias_RMS	0xFA 0x03	0x04 0x00	float											
Accels_noise_rms	0x06 0x04	0x04 0x00	float											
Accel_dynamic_gain	0x0A 0x04	0x04 0x00	float											
Dynamic_LPF	0x0E 0x04	0x04 0x00	float											
Align_gyro_STD_threshold	0x14 0x04	0x04 0x00	float	This parameter affects the initialization algorithm (see section 6.7). <table><tr><th>State</th><th>Value</th></tr><tr><td>Disabled (default)</td><td>0x00</td></tr><tr><td>Enabled</td><td>0x01</td></tr></table>	State	Value	Disabled (default)	0x00	Enabled	0x01				
State	Value													
Disabled (default)	0x00													
Enabled	0x01													
ZUPT	0x13 0x04	0x01 0x00	byte											
2.2. Filtering														
Filter	0x51 0x04	0x01 0x00	byte	A cascade of filters for filtering of the transmitted accelerometers and gyroscopes signals. <table><tr><th>Filter</th><th>State</th><th>Mask</th></tr><tr><td>Filter1</td><td rowspan="3">0 – Disabled 1 – Enabled</td><td>0x01</td></tr><tr><td>Filter2</td><td>0x02</td></tr><tr><td>Filter3</td><td>0x04</td></tr></table> Example 1:	Filter	State	Mask	Filter1	0 – Disabled 1 – Enabled	0x01	Filter2	0x02	Filter3	0x04
Filter	State	Mask												
Filter1	0 – Disabled 1 – Enabled	0x01												
Filter2		0x02												
Filter3		0x04												

				Filter1 (Enabled) + Filter2 (Enabled) + Filter3 (Disabled) = 0x03. Example 2: Filter1 (Enabled) + Filter2 (Enabled) + Filter3 (Enabled) = 0x07.														
Filt1_Dzeta 1	0x52 0x04	0x04 0x00	float	Parameters for setting the device's cascade filter (see section 6.8).														
Filt1_Dzeta 2	0x56 0x04	0x04 0x00	float															
Filt1_W	0x5A 0x04	0x04 0x00	float															
Filt2_Dzeta 1	0x5E 0x04	0x04 0x00	float															
Filt2_Dzeta 2	0x62 0x04	0x04 0x00	float															
Filt2_W	0x66 0x04	0x04 0x00	float															
Tau	0x6A 0x04	0x04 0x00	float															
3. Sensors options																		
3.1. Settings																		
LPF_Bandwidth	0xD7 0x03	0x01 0x00	byte	This specifies the bandwidth of the low-pass filter (LPF) which is applied to the sensors output data. <table><tr><th>Bandwidth</th><th>Value</th></tr><tr><td>16 Hz</td><td>0x00</td></tr><tr><td>33 Hz</td><td>0x01</td></tr><tr><td>66 Hz</td><td>0x02</td></tr><tr><td>131 Hz</td><td>0x03</td></tr><tr><td>262 Hz</td><td>0x04</td></tr><tr><td>Disabled</td><td>0xFF</td></tr></table>	Bandwidth	Value	16 Hz	0x00	33 Hz	0x01	66 Hz	0x02	131 Hz	0x03	262 Hz	0x04	Disabled	0xFF
Bandwidth	Value																	
16 Hz	0x00																	
33 Hz	0x01																	
66 Hz	0x02																	
131 Hz	0x03																	
262 Hz	0x04																	
Disabled	0xFF																	
Output_variant	0xD0 0x03	0x01 0x00	byte	Type of gyros and accelerometers output data (see section 6.3.6). <table><tr><th>Type</th><th>Value</th></tr><tr><td>Instant (default)</td><td>0x00</td></tr><tr><td>Average</td><td>0x01</td></tr><tr><td>Incremental</td><td>0x02</td></tr></table>	Type	Value	Instant (default)	0x00	Average	0x01	Incremental	0x02						
Type	Value																	
Instant (default)	0x00																	
Average	0x01																	
Incremental	0x02																	
IMU-P_Interface	0x96 0x04	0x01 0x00	byte	Reserved (not supported in KERNEL devices).														
Min_Vinp_threshold	0xDA 0x03	0x02 0x00	word	Reserved (not supported in KERNEL devices).														
Max_Vinp_threshold	0xDC 0x03	0x02 0x00	word															
Sensors measurement range (read only)	0x8F 0x02	0x01 0x00	byte	This parameter allows the user to see the measurement ranges of the device's gyroscopes and accelerometers. The four least significant bits are the accelerometer range, and the four most significant bits are the gyroscope range. The values for gyroscopes: <table><tr><th>Hex</th><th>Bin</th><th>Gyros range</th></tr><tr><td>6</td><td>0110</td><td>450 deg/sec</td></tr><tr><td>8</td><td>1000</td><td>950 deg/sec</td></tr></table>	Hex	Bin	Gyros range	6	0110	450 deg/sec	8	1000	950 deg/sec					
Hex	Bin	Gyros range																
6	0110	450 deg/sec																
8	1000	950 deg/sec																

				9	1001	2000 deg/sec
				The values for accelerometers:		
				Hex	Bin	Accel. range
				3	0011	8 g
				5	0101	15 g
				6	0110	40 g
				For an example of decoding such a byte, refer to section 6.3.8.1.		

Field 4 (Check sum)

“Check sum” = sum of bytes starting from byte #2 of the whole packet to the last byte of the payload.

6.3.7.1. Command Use Examples

Example 1 (Data_rate change)

The SaveFlash command general structure:

AA 55 00 00 0E 00 B2 FF 12 00 02 00 **RT RT CS CS**

where **RT RT** is rate (2 bytes, word); **CS CS** – check sum.

To set rate as 2000Hz, the SaveFlash command looks as follows:

AA 55 00 00 0E 00 B2 FF 12 00 02 00 **D0 07 AA 02**

Example 2 (Initial_alignment_time change)

The SaveFlash command general structure:

AA 55 00 00 0E 00 B2 FF 14 00 02 00 **IA IA CS CS**

Where **IA IA** is initial alignment time (2 bytes, word); **CS CS** – check sum.

To set time as 15 seconds, the SaveFlash command looks as follows:

AA 55 00 00 0E 00 B2 FF 14 00 02 00 **0F 00 E4 01**

Example 3 (Alignment_angles change)

The SaveFlash command general structure:

AA 55 00 00 18 00 B2 FF 3A 02 0C 00 **A1 A1 A1 A1**
A2 A2 A2 A2 A3 A3 A3 A3 CS CS

where AN AN AN AN - alignment angles (4 bytes, float); **CS CS** – check sum.

To set angles as 10.0, 20.0, and 30.0 degrees, the SaveFlash command looks as follows:

AA 55 00 00 18 00 B2 FF 3A 02 0C 00 **00 00 20 41**
00 00 A0 41 00 00 F0 41 84 04

Note: 10.0 (dec) = 41 20 00 00 (float)
20.0 (dec) = 41 A0 00 00 (float)
30.0 (dec) = 41 F0 00 00 (float)

Example 4 (LPF Bandwidth change)

The SaveFlash command general structure:

0xAA 0x55 0x00 0x00 0x0D 0x00 0xB2 0xFF 0xD7 0x03 0x01 0x00
ID CS CS

Where ID is the code of the bandwidth, **CS CS** – check sum.

For example, to set filter to 262 Hz bandwidth, the following command should be used:

0xAA 0x55 0x00 0x00 0x0D 0x00 0xB2 0xFF 0xD7 0x03 0x01 0x00
0x04 0x9D 0x02

6.3.8. ReadRAM Command

The ReadRAM command allows the user to read the data of individual parameter in the IMU RAM memory.

The list of available parameters is shown in Table 6.23. The ReadRAM command structure corresponds to the SaveFlash command and can be seen in Table 6.21, with payload shown in Table 6.24.

The device responds to the ReadRAM command with a message containing the parameter value in its payload. The byte structure of this message is shown in Table 6.2

Table 6.24. The ReadRAM command Payload structure

Field #	3.1	3.2	3.3
Bytes Offset	0 – 1	2 – 3	4 – 5
Data type	word	word	word
Size	2	2	2
Description	Command ID	Parameter ID	Parameter length
Value	0xB1 0xFF	Variable (see Table 6.23)	Variable (see Table 6.23)

6.3.8.1. Command Use Examples

Example 1 (Data rate read)

The ReadRAM command general structure:

AA 55 00 00 0C 00 B1 FF **12 00 02 00 D0 01**

Where **12 00** is a parameter ID for Data_Rate; **02 00** - a parameter length for Data Rate; **D0 01** - a check sum.

Example 2 (Initial alignment time read)

The ReadRAM command general structure:

AA 55 00 00 0C 00 B1 FF **14 00 02 00 D2 01**

Where **14 00** is a parameter ID for Initial_alignment_time; **02 00** - a parameter length for Initial alignment time; **D2 01** - a check sum.

Example 3 (Sensors measurement range read)

The ReadRAM command general structure:

AA 55 00 00 0C 00 B1 FF 8F 02 01 00 4E 02

Where 8F 02 is the parameter ID, 01 00 is the parameter length, and 4E 02 is the check sum.

The highlighted payload of the response message includes information about the measurement ranges:

AA 55 01 B1 07 00 93 4C 01

Where 9 is a gyro range of ± 2000 dps and 3 represents an accelerometer range of ± 8 g.

Refer to Table 6.23 to see the values of other possible ranges.

6.4. The Unit Status Word definition

The Unit Status Word (USW) provides information on the current state of the KERNEL IMU. Status of each bit of the USW warning byte is specified in the Table 6.25.

Table 6.25. The Unit Status Word description

Byte	Bit	Parameter	Description
Low byte	0	Reserved	
	1	Internal sensors communication	0 – No failure 1 – Failure detected
	2	Internal sensors configuration	0 – No failure 1 – Failure detected
	3-7	Reserved	
High byte	8	Reserved	
	9	Reserved	
	10	Angular Rate Exceeding Detect	0 – X-angular rate is within the range 1 – X-angular rate is out of range
	11		0 – Y-angular rate is within the range 1 – Y-angular rate is out of range
	12		0 – Z-angular rate is within the range 1 – Z-angular rate is out of range
	13	Reserved	
	14	Environmental Temperature	0 – Temperature is within the operating range 1 – Temperature is out of the operating range
	15	Reserved	

For firmware from 20-Aug-2021 and newer, the parameter USW Extended is available for output. The status of each bit of the USW Extended is specified in Table 6.26.

Table 6.26. The Unit Status Word Extended description

Byte	Bit	Parameter	Description
Low byte	0	Initial alignment status	0 – initial alignment in process 1 – initial alignment completed
	1	Gyro bias estimation	0 – Unsuccessful 1 – Successful
	2-7	Reserved	
High byte	8-15	Reserved	

6.5. Limitation of the IMU Maximum Output Data Rate

When configuring the output data rate for the KERNEL IMU using SaveFlash command (see section 6.3.7, respectively), it is essential to ensure that the chosen COM port baud rate is capable of handling the chosen data rate throughput. The maximum data rate (Hz) can be calculated using baud rate and data package length:

$$\text{max_data_rate} = \frac{\text{COM_baud_rate}}{\text{bits_per_byte} * \text{package_length}} \quad (6.1)$$

where COM_baud_rate is the COM port baud rate (bits/s); bits_per_byte = 11 bits per one transferred byte of data; package_length for binary data = payload length plus 8 bytes of overhead. See Table 6.4 through Table 6.14 for payload length of binary output data formats. The package_length of text output data formats corresponds to their structure.

The KERNEL can only output at data rates that are factors of 2000 (1, 2, 4, 5, 8, 10, 16, 20, 25, 40, 50, 80, 100, 125, 200, 250, 400, 500, 1000, 2000 Hz). Calculated maximum data rates using equation (6.1) must be rounded down to the nearest factor of 2000 that is listed above. The actual values of the maximum data rates calculated in this way for each output data format shown in Table 6.27.

Table 6.27. KERNEL maximum data rate examples for different output data formats

Output data format	Data package length, bytes	COM-port baud rate, bps			
		38400	115200	460800	2000000
		Maximum data rate, Hz			
KERNEL Orientation	34+8	80	200	500	2000
KERNEL GA Data	32+8	80	250	1000	2000
KERNEL GAm Data	32+8	80	250	1000	2000
KERNEL Quaternion Data	34+8	80	200	500	2000
KERNEL Calibrated HR Data	52+8	50	125	500	2000
KERNEL GAA Data	44+8	50	200	500	2000
KERNEL GAAM Data	44+8	50	200	500	2000

Notes:

1. The 1843200, 3686400, and 4000000 baud rate values were designed for specific purposes only, and a particular adaptor for data/commands transmission is required. Do not apply such values to the IMU without Inertial Labs agreement.
2. If the default baud rate for the IMU COM port is changing, then the same baud rate must also be set for the COM port of the host computer to keep data/commands transmission.
3. The list of supported baud rates may vary from one device to another device. The supported ones depend on the user's needs. In case of any questions related to the baud rate change, please contact the Inertial Labs support team.

6.6. Finite Impulse Response (FIR) Filter Delay

The KERNEL IMU has options implemented for the user to enable or disable digital filtering of the transmitted signals. Inertial Labs uses a Low Pass Filter (LPF) which is more specifically a FIR filter. The FIR filter delay is dependent on two things, the bandwidth of the filter signal and the filter order number. Table 6.28 shows these values tabulated.

Table 6.28. Tabulated values for the FIR filter delay.

Filter Bandwidth (Hz)	Filter Order (N)	Delay (Seconds)
16	192	0.048
33	96	0.024
66	48	0.012
131	24	0.006
262	12	0.003

6.7. Initial Alignment Logic (ZUPT Algorithm)

The initial alignment process includes the KERNEL IMU estimating gyro biases. Gyro bias is estimated when the initial alignment is set to value ≥ 1 second. Gyro bias estimation includes the Earth rotation component.

There is the ZUPT parameter which affects the initialization algorithm in the following ways:

- ZUPT is disabled

When ZUPT is disabled, then gyro bias is estimated via averaging. In the IMU parameters, the alignment gyro STD threshold is set that is used to determine if the estimated during the alignment gyro bias is accepted or rejected (zeroed). The object must remain static with no vibration during the initial alignment when ZUPT is disabled.

- ZUPT is enabled

When ZUPT is enabled, then the STD threshold is not used and algorithm is using KF for bias estimation. The object must remain static but some low-level vibration is acceptable for this type of initial alignment.

Note: The initial alignment time, alignment gyro STD threshold, and ZUPT parameter can be configured using SaveFlash command. To do this, you need to modify the Initial_alignment_time, Alignment_gyro_STD_threshold, or ZUPT parameters (see

Table 6.23). Alternatively, you can use the IMU GUI software to configure IMU (refer to the User Manual, section 4.5.3.1 for details).

The IMU unit may report initial alignment status and the status of gyro bias estimation calculation based on standard deviation measurement. In this order, USW Extended data should be chosen for output through UDD (see section 6.2.8 and 6.4).

6.8. Cascade Filter

The KERNEL IMU has implemented a cascade of filters for filtering of the transmitted accelerometers and gyroscopes signals. The cascade filter includes a combination of the first and second orders IIR filters. The filter structure is defined in Figure 6.1.

Note: This cascade filter was designed for specific operations only and for experienced users. For questions on this filter, please contact the Inertial Labs Support Team.

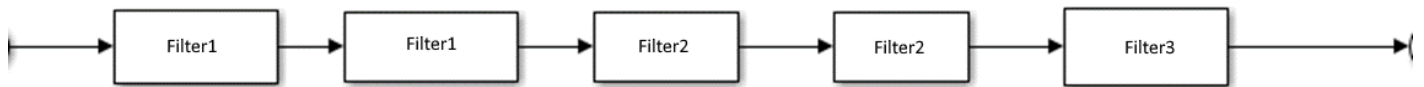


Figure 6.1. IMU output filter cascade.

Filter1 and Filter2 are the second order filters with transfer function:

$$Filter_1(s) = \frac{s^2 + 2\zeta_{11}\omega_1s + \omega_1^2}{s^2 + 2\zeta_{21}\omega_1s + \omega_1^2} \qquad Filter_2(s) = \frac{s^2 + 2\zeta_{12}\omega_2s + \omega_2^2}{s^2 + 2\zeta_{22}\omega_2s + \omega_2^2}$$

Filter3 is the first order filter with transfer function:

$$Filter_3(s) = \frac{1}{\tau s + 1}$$

$Filter_1 * Filter_1$, $Filter_2 * Filter_2$, or $Filter_3$ can be configured by the user so that can be bypassed or activated. Moreover, parameters ζ_{11} , ζ_{21} , ω_1 , ζ_{12} , ζ_{22} , ω_2 , and τ can be modified. The filtering option can be enabled, modified and disabled using SaveFlash command. To do this you need to modify Filter parameter (see Table 6.23). All these modifications are supported by the Inertial Labs IMU GUI software. For details on that, refer to the IMU GUI User Manual, section "4.5.3.2 Additional IMU Filtering".

The result of signal filtering can be seen outputting the following data through UDD (see section 6.2.8 for that):

- "Filtered low-G accels HR" (id 0x91)
- "Filtered high-G accels HR" (id 0x92)
- "Bias-compensated gyro data" (id 0x2D).

7. IMU Automatic Start

The auto start mode allows the unit to start operation and data output after power is applied without any additional commands from the host computer. It is possible to choose the desirable output data format for auto start (see section 6.2).

The auto start option can be enabled or disabled using the SaveFlash command. To do this, modify the Auto_Start parameter (see Table 6.23). Alternatively, use the IMU GUI software to configure the device. Refer to the User Manual Section 4.5.2 for more details.

If auto start mode is enabled and power is available, the device automatically starts by sending the message AA 55 01 00 08 00 00 00 09 00 (hexadecimal format) as an acknowledgment and then begins data output according to the chosen data format.

To stop the KERNEL, send the Stop command (see Section 6.3.2). After receiving the Stop command, the KERNEL stops data calculation, goes into idle mode, and remains ready to receive any command from the host computer.

Appendix A. Optional KERNEL Device with Ethernet Interface

A.1. Connection Overview

The Inertial Labs™ device is delivered to a customer with a multi-conductor cable. One side of the cable should be connected to the socket on the rear panel of the device. The other side of the cable is attached to the breakout board with numerous connectors on it (see Figure A.1 and Figure A.2). The connectors needed for Ethernet communication are XS6 (used for a data) and XS7 (used for a power). The device with Ethernet retains the communication via a serial port as well.

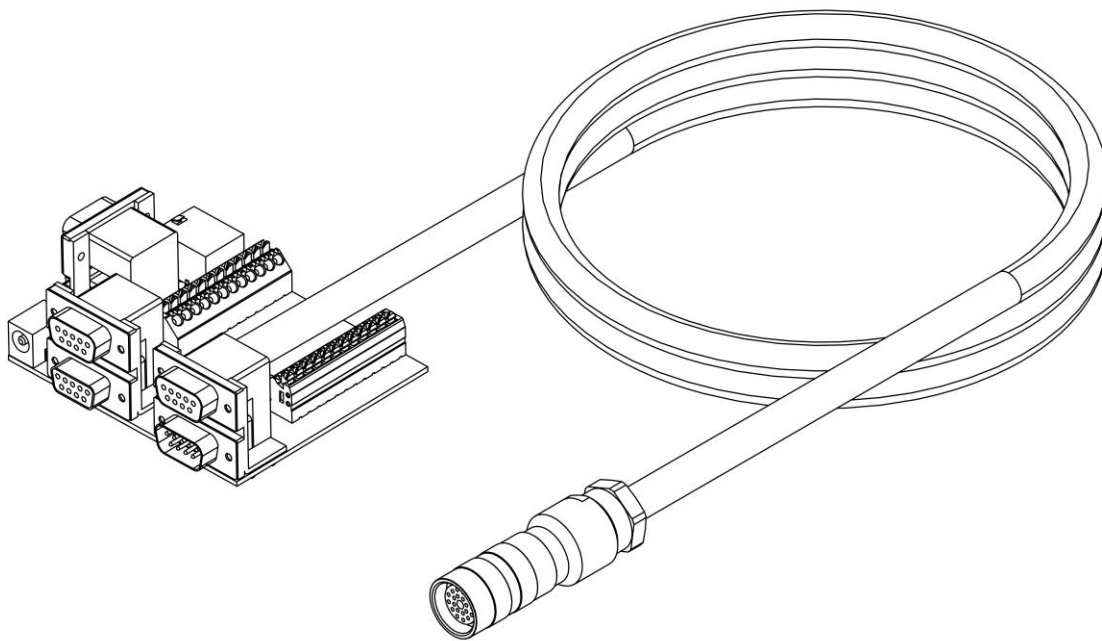


Figure A.1. Multi Interface cable with breakout board.

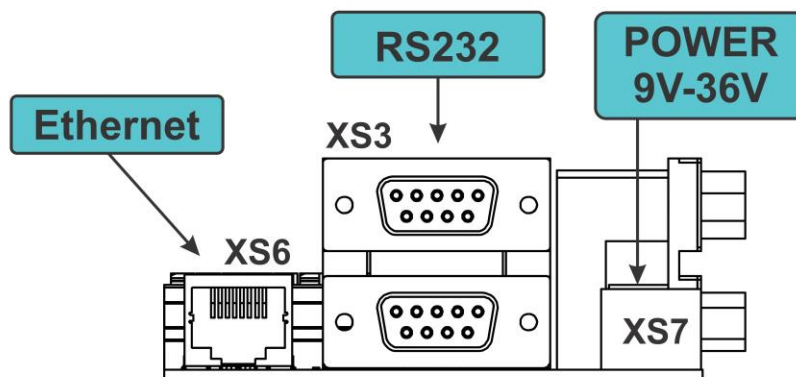


Figure A.2. Multi Interface breakout board functionality.

By a specific request the device can be equipped with the cable assembly that provides Ethernet and power via a single RJ45 connector. For the testing purposes the PoE adapter (so called PoE injector) is also supplied. The example of this cable is shown on the picture below (see Figure A.3).

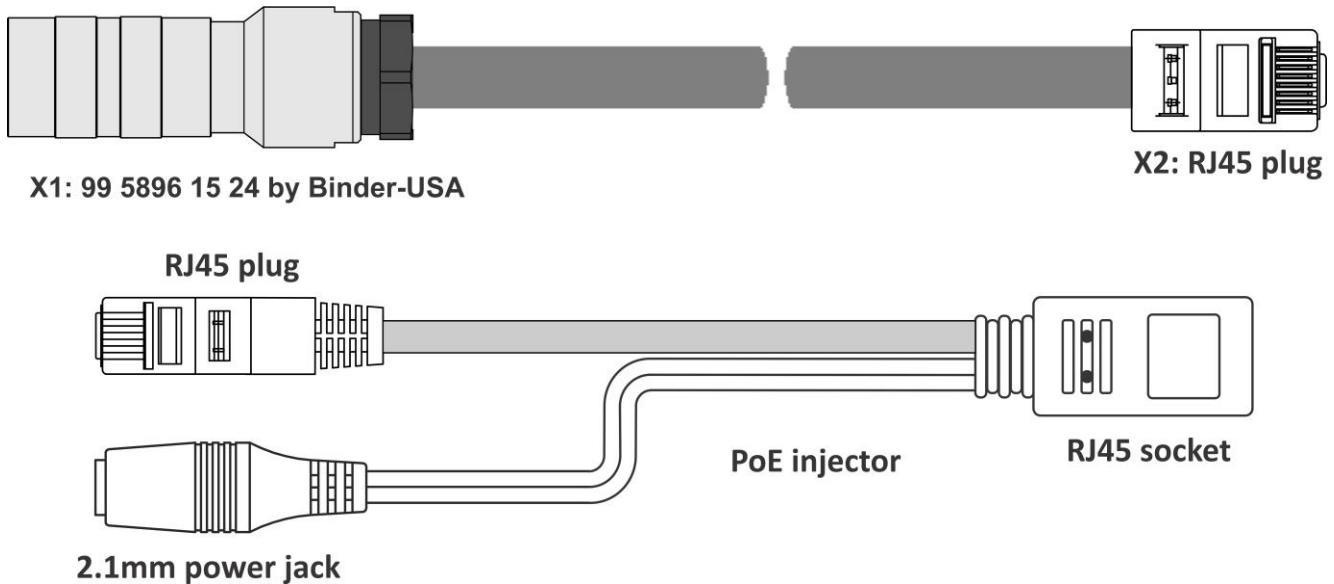


Figure A.3. Single port cable and PoE injector.

A.2. Connection Steps

Step 1.

Attach the cable to the connector on the rear wall of the device.

Step 2.

Connect the breakout board to your local network via XS6 socket using the Ethernet patch cord, or connect the RJ45 plug of the PoE injector to your local network.

Step 3.

Connect the barrel power plug to the XS7 socket on the breakout board, or to the barrel socket on the PoE injector. The LED on the device should light up yellow and then turn to red (note 1). The device is ready to operate.

Note 1: As soon as the Ethernet cable is plugged to the LAN the LED starts switching from red to yellow when there is an activity on the link, so it acts as an Activity LED of the Ethernet port.

A.3. Networking Details

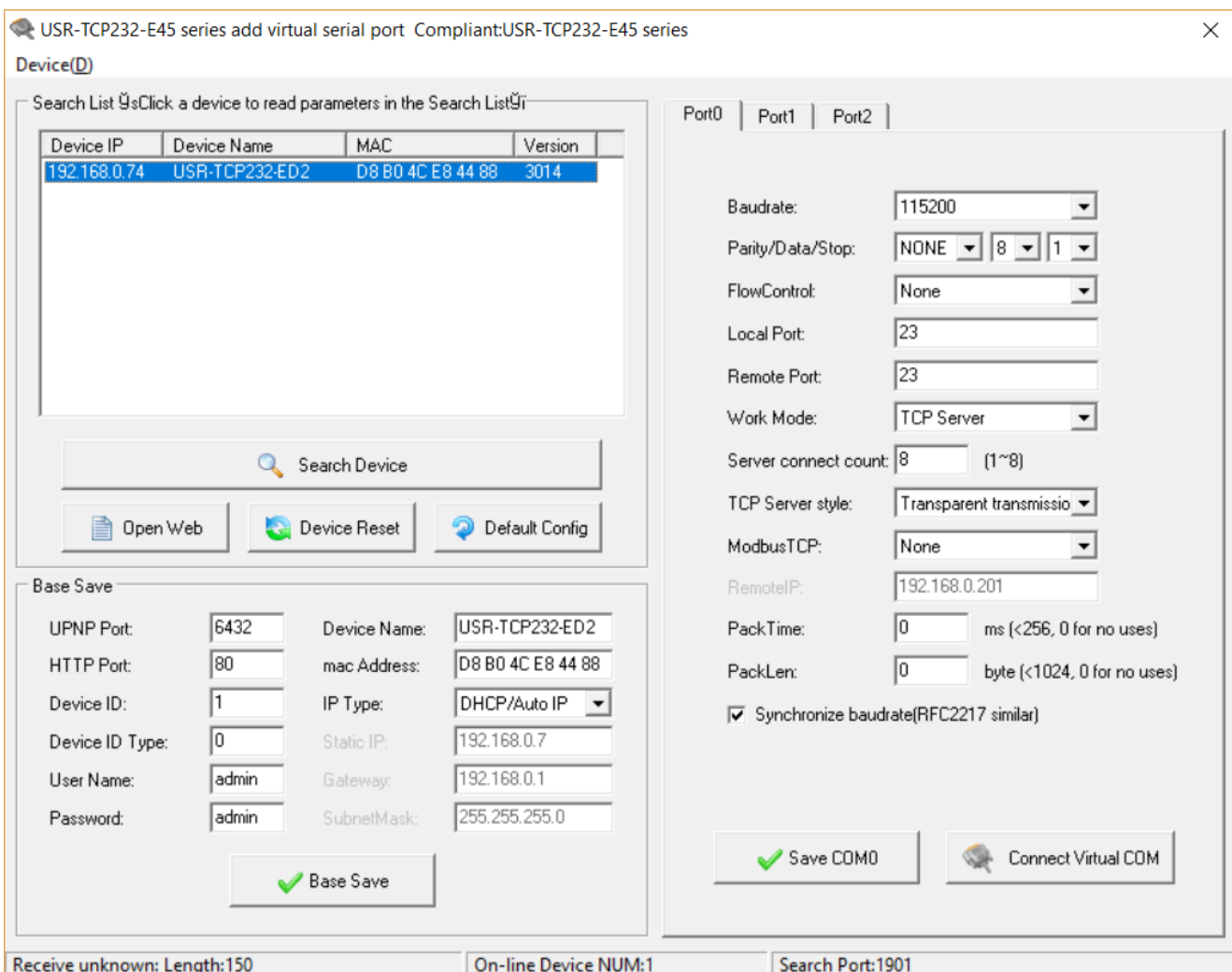
The Ethernet module inside of the device is preconfigured for obtaining an IP address automatically. In most cases the IP address is assigned immediately after the device is powered and connected to the LAN. To troubleshoot please access your router console and find the "DHCP" page. Make sure the DHCP server is ON with the MAC address filtering feature OFF or add the MAC address of the unit to the list of "known devices". Then locate the DHCPClient tab and check the list of connected devices for the IP that was assigned to the device.

A.4. Checking the Connection to the Network

After the unit has obtained an IP address it becomes accessible via the local network. To check the accessibility and re-configure the Ethernet module (if needed) please install the USR-VCOM software to the PC connected to the same network.

The USR-VCOM is developed by USR-IOT for configuring the Ethernet module and checking its accessibility via the local network. It can also create a virtual serial port on the PC connected to the same network. It can be downloaded from here: <https://www.pusr.com/support/downloads/usr-vcom-setup-software.html>

Run the software and select Tools -> Search -> USR-TCP232-E45. You will see the device in the “Search List” as shown in Figure A.4. You can assign fixed network parameters like Static IP, Subnet Mask and Gateway using this tool. Do so or leave all settings as is.



USR-TCP232-E45 series add virtual serial port Compliant:USR-TCP232-E45 series

Device(D)

Search List Click a device to read parameters in the Search List

Device IP	Device Name	MAC	Version
192.168.0.74	USR-TCP232-ED2	D8 B0 4C E8 44 88	3014

Search Device

Open Web Device Reset Default Config

Base Save

UPNP Port: 6432 Device Name: USR-TCP232-ED2

HTTP Port: 80 mac Address: D8 B0 4C E8 44 88

Device ID: 1 IP Type: DHCP/Auto IP

Device ID Type: 0 Static IP: 192.168.0.7

User Name: admin Gateway: 192.168.0.1

Password: admin SubnetMask: 255.255.255.0

Base Save

Port0 Port1 Port2

Baudrate: 115200

Parity/Data/Stop: NONE 8 1

FlowControl: None

Local Port: 23

Remote Port: 23

Work Mode: TCP Server

Server connect count: 8 (1~8)

TCP Server style: Transparent transmissio

ModbusTCP: None

RemotelP: 192.168.0.201

PackTime: 0 ms (<256, 0 for no uses)

PackLen: 0 byte (<1024, 0 for no uses)

☒ Synchronize baudrate(RFC2217 similar)

Save COM0 Connect Virtual COM

Receive unknown: Length:150 On-line Device NUM:1 Search Port:1901

Figure A.4. The USR-VCOM search window shows an Ethernet device found in the LAN.

- Note 2: USR-VCOM uses broadcast messages when scanning the network for the Ethernet Module. Please make sure the broadcast messages are not blocked in your LAN.
- Note 3: Unless otherwise is requested, the device is supplied to the customer pre-configured to obtain the IP address dynamically and with the Port 0 configured as TCP Server, i.e., ready for the Virtual Serial Port. No configuration is required.

A.5. Preface

This appendix describes the basic principles of communication with the Inertial Labs device which have a built-in Ethernet port. Such device is equipped with a USR-TCP232-ED2 module made by USR-IOT. This document refers to the website of USR-IOT as to the location of required software tools and datasheets, <https://www.pusr.com/products/ethernet-modules>

A.6. Communication VIA Ethernet USING a Virtual Serial Port

It is possible to communicate with the device using the virtual serial port. In order to create a virtual serial port, please go to the USR-VCOM search window as directed in the previous clause; then choose the desired module in the Search List Window (see Figure A.4.). Then click Connect Virtual COM on the bottom of the Port0 tab. In the new window choose the serial port number (can be any number not occupied by another hardware on your PC) and add the optional description of the port. Leave the other fields as shown of the Figure A.4 and Figure A.5.

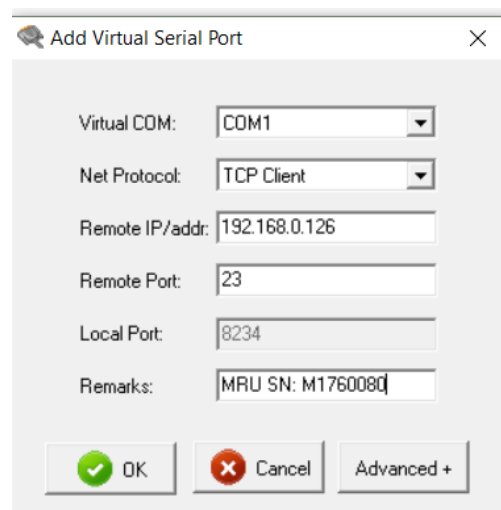


Figure A.5. Add Virtual Serial Port (VSP) window.

Press “OK” and you will see the main window of USR-VCOM with the VSP created.

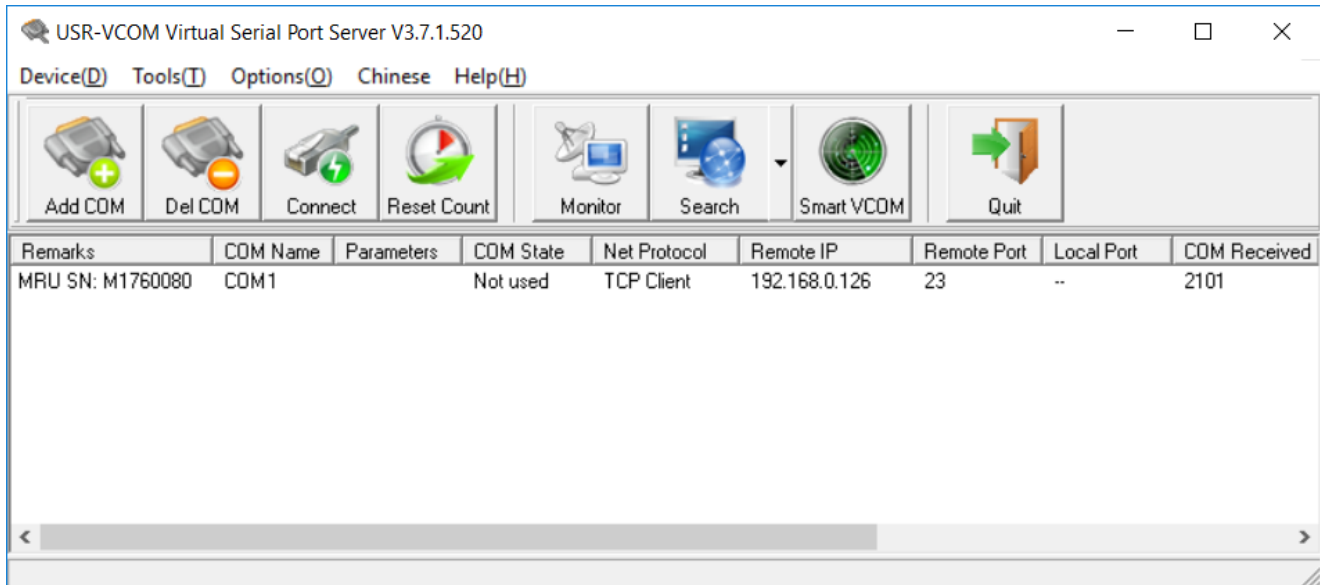


Figure A.6. Virtual Serial Port COM1 is created for the Port0 of the 192.168.0.126

Now you can use this virtual serial port for communication with the device.

For KERNEL: use the Inertial Labs IMU GUI. Please refer to the Inertial Labs IMU GUI User's Manual.

A.7. Communication OVER the Ethernet IN TCP/IP MODE

It is possible to communicate with the device via the Ethernet directly without using any software from Inertial Labs. To establish the connection and send commands to the device it is necessary to perform the following steps:

Step 1.

Use the USR-VCOM software to inquire the IP address and the number of the Local Port. On the example below the IP address is static and equals to 192.168.0.7, Local Port number is 23, see the Figure A.7.

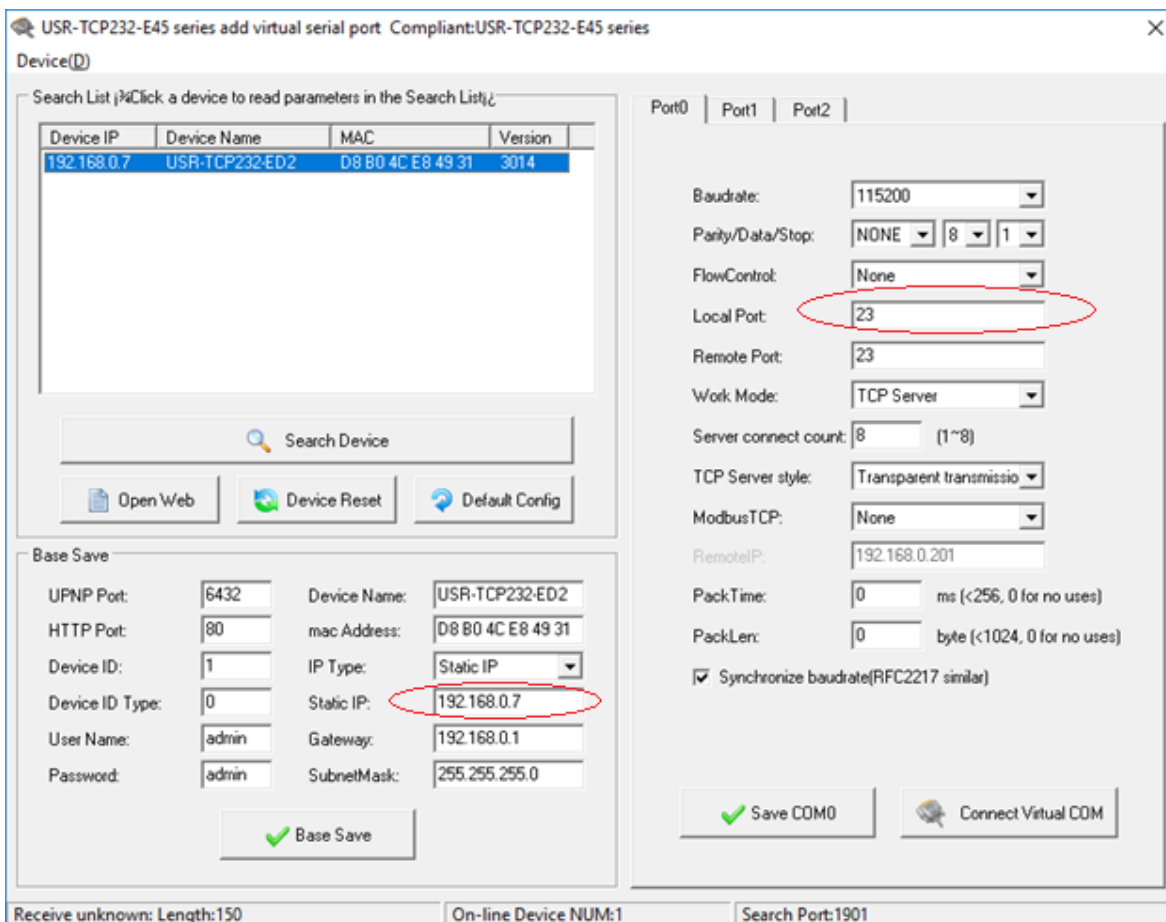


Figure A.7. The USR-VCOM search window shows an Ethernet device found in the LAN.

Please note, that Baudrate value in the Port0 tab should match the Baudrate of the main port of the device. If the value is different - please correct and press Save COM0. (see Figure A.7).

Note 4: For KERNEL: create the VSP, run the IL Sensor Configuration Tool, choose the port with option “Scan All Available” and press Start Scanning. This will perform the scan on all available baudrates and report the current one.

Step 2.

To send commands through the Ethernet any TCP terminal can be used. This document uses the RealTerm tool as an example. In the “Port” tab of the RealTerm it is necessary to set the IP address and Local port number obtained on the Step 1 in the following format: **ip_address:port_number** (please see the Figure A.8). After the IP address and Local port number are set, please click “Open” button.

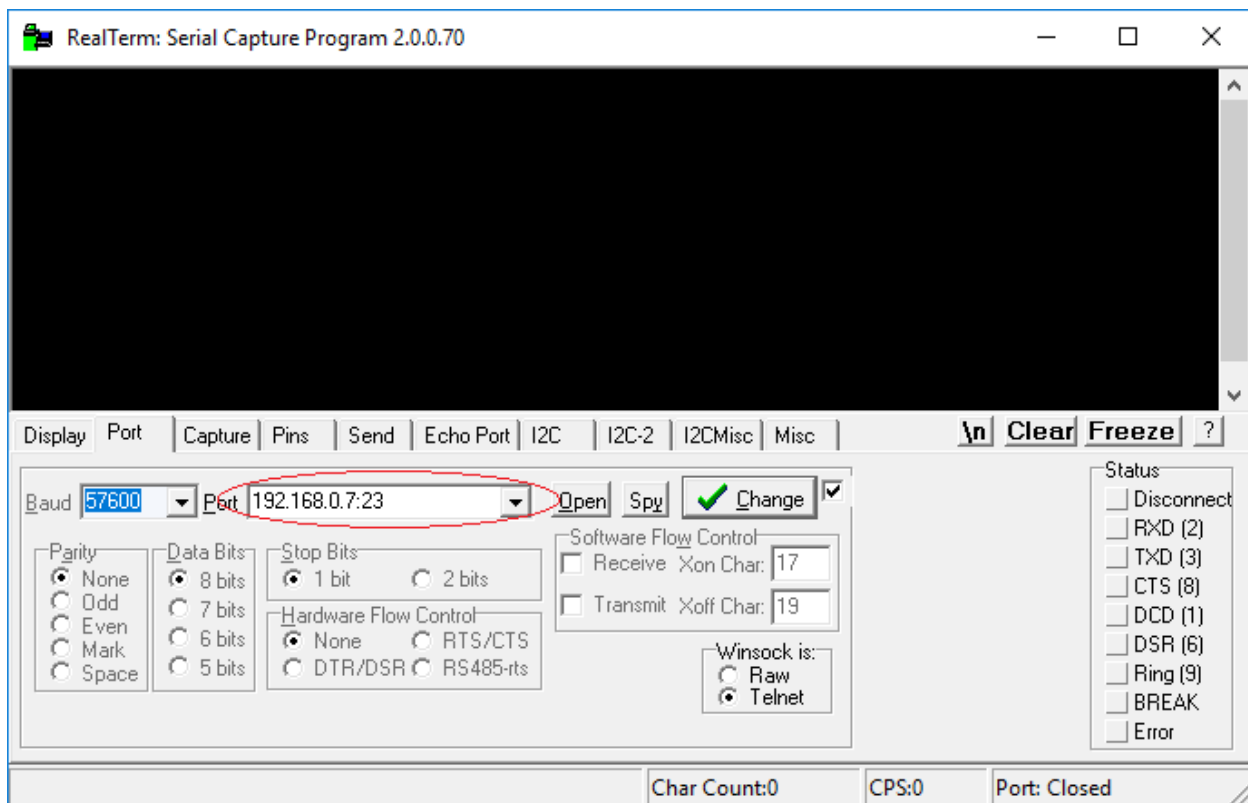


Figure A.8. The RealTerm tool, port settings.

Step 3.

This example demonstrates how to pull a data message by message.

On the “Send” tab enter the command in hexadecimal format and then press “Send numbers” button. On the Figure A.9 the request to KERNEL to return the Euler angles is shown as an example. The requested data appears on the window each time the “Send numbers” button is pressed.

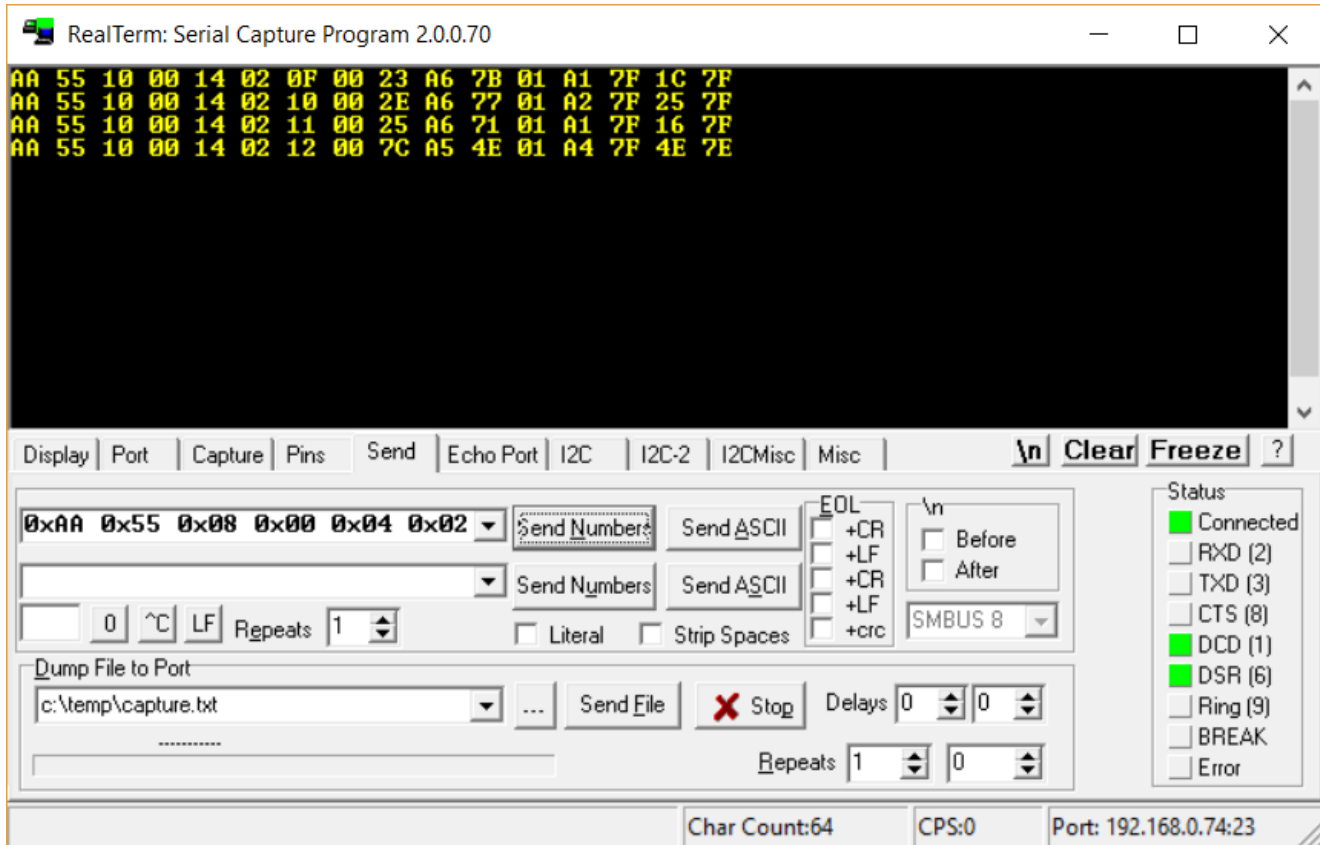


Figure A.9. The Euler angles requested and returned from KERNEL.

The next example demonstrates how to start the data stream from a device.

In order to start the data stream, it is necessary to send the required command to the device (see “Appendix B. Full List of the IMU Commands”).

Enter the message in the hexadecimal format to the “Send” tab of the RealTerm tool. Then click “Send Numbers” button. When the command is received by the device it may start the initial alignment process (only if the device is configured to output orientation or quaternion angles). After the initial alignment procedure is over, the device starts transmitting data. If the initial alignment time is set to 0, then the initial alignment procedure is skipped and device starts the data output without any pause.

Step 4.

To stop the data stream, the Stop command should be sent to the device in the same way as described in Step 3.

A.8. Communication Over the Ethernet in Auto Transfer Mode

When the device is configured to output data after power up without any commands from the host computer (auto transfer mode) the Ethernet should be configured to work in the UDP mode. For this purpose, please perform the following steps:

Step 1.

Please configure the device to auto transfer mode. The main settings of auto transfer mode are the baudrate of the serial port to which COM1 of the device is connected, output data rate, and output format.

Step 2.

Open the USR-VCOM tool and go to Tools -> Search -> USR-TCP232-E45. Click on the device that appeared in the "Search List" as shown in Figure A.4.

Step 3.

In the field 'Baudrate' please set the value that corresponds to the baudrate set on the Step 1. Please note that maximum possible baudrate is 2000000 bps. Please set the 'Work Mode' parameter to 'UDP' and specify the IP address of the data receiver in the 'Remote IP' field (see Figure A.10). Click 'Save COM0' button to save the settings.

Step 4.

Power-cycle the device to start automatic data output via the Ethernet port.

Step 5.

To stop data stream power off the device.

USR-TCP232-E45 series add virtual serial port Compliant:USR-TCP232-E45 series

Device(D)

Search List | Click a device to read parameters in the Search List |

Device IP	Device Name	MAC	Version
192.168.0.77	USR-TCP232-ED2	D8 B0 4C E8 44 78	3014

Search Device

Open Web Device Reset Default Config

Base Save

UPNP Port: 6432 Device Name: USR-TCP232-ED2

HTTP Port: 80 mac Address: D8 B0 4C E8 44 78

Device ID: 1 IP Type: DHCP/Auto IP

Device ID Type: 0 Static IP: 192.168.0.7

User Name: admin Gateway: 192.168.0.1

Password: admin SubnetMask: 255.255.255.0

Base Save

Port0 Port1 Port2

Baudrate: 230400

Parity/Data/Stop: NONE 8 1

FlowControl: None

Local Port: 23

Remote Port: 23

Work Mode: UDP

Server connect count: 8 (1~8)

TCP Server style: Transparent transmissio

ModbusTCP: None

RemoteIP: 192.168.0.30

PackTime: 0 ms (<256, 0 for no uses)

PackLen: 0 byte (<1024, 0 for no uses)

☒ Synchronize baudrate(RFC2217 similar)

Save COM0 Connect Virtual COM

Receive unknown: Length:150 On-line Device NUM:1 Search Port:1901

Figure A.10. Configuring the Ethernet module to UDP mode.

Appendix B. Full List of the IMU Commands

All the KERNEL commands have a byte structure as shown in the Table 6.2. Payload for all commands has length 1 byte and contains the code of a command. All commands and their exact structure in hexadecimal numbers are listed below in Table B.1.

Table B.1. List of the KERNEL commands with exact structure

Command name	Code	Exact structure (hex)
Commands for Inertial Labs™ KERNEL control		
KERNEL_Orientation	0x33	AA 55 00 00 07 00 33 3A 00
KERNEL_GAData	0x8F	AA 55 00 00 07 00 8F 96 00
KERNEL_GAmData	0x9B	AA 55 00 00 07 00 9B A2 00
KERNEL_GAADData	0xA5	AA 55 00 00 07 00 A5 AC 00
KERNEL_GAAmData	0xA6	AA 55 00 00 07 00 A6 AD 00
KERNEL_QuatData	0x82	AA 55 00 00 07 00 82 89 00
KERNEL_CalibHR	0x81	AA 55 00 00 07 00 81 88 00
Stop	0xFE	AA 55 00 00 07 00 FE 05 01
User_Def_Data	0x95	AA 55 00 00 07 00 95 9C 00
User_Def_Data_config	0x96	AA 55 00 00 07 00 96 9D 00
Get_User_Def_Data_struct	0x97	AA 55 00 00 07 00 97 9E 00
GetDevInfo	0x12	AA 55 00 00 07 00 12 19 00
GetBIT	0x1A	AA 55 00 00 07 00 1A 21 00
SaveFlash	–	Has a variable structure, see section 6.3.7
ReadRam	–	Has a variable structure, see section 6.3.8

Appendix C. Variants of the IMU Mounting Relative to the Object Axes

The KERNEL IMU can be mounted on the object in any known position (upside-down, backwards, upright, etc.) relative to the object axes. Such mounting doesn't change the validity of calculations of carrier object orientation angles as long as the KERNEL's mounting offset angles are correctly stored in the KERNEL's nonvolatile memory.

In order to set mounting angles into the IMU-P nonvolatile memory, please use the SaveFlash command. To do this you need to modify the Alignment_angles parameter (see Table 6.23). Alternatively, you can use Inertial Labs IMU GUI (refer to the User Manual, section 4.5.2 for details).

Angles of the KERNEL position (alignment angles) are set in following order (like heading, pitch and roll setting) in degrees:

- The first alignment angle (heading) is determined by IMU rotation around its Z-axis. Clockwise rotation is denoted as positive. The range for this alignment angle is -180° to $+180^{\circ}$.

In other words, this angle sets the misalignment between the longitudinal axes of the IMU and the carrier object in the horizon plane.

- The second alignment angle (pitch) is determined by IMU rotation around its X-axis. Counterclockwise rotation is denoted as positive. The range for this alignment angle is -90° to $+90^{\circ}$.

In other words, this angle sets the inclination of the longitudinal axis of the IMU relative to the carrier object horizon plane.

- The third alignment angle (roll) is determined by IMU rotation around its Y-axis. Counterclockwise rotation is denoted as positive. The range for this alignment angle is -180° to $+180^{\circ}$.

Note: Default values of the IMU alignment angles are all zeros.

Check the validity of the entered alignment angles by using the visualization tab in the IMU GUI (further discussed in IMU GUI User Manual).

Some examples of the Inertial Labs KERNEL mounting relative the carrier object is shown in Figure C.1.

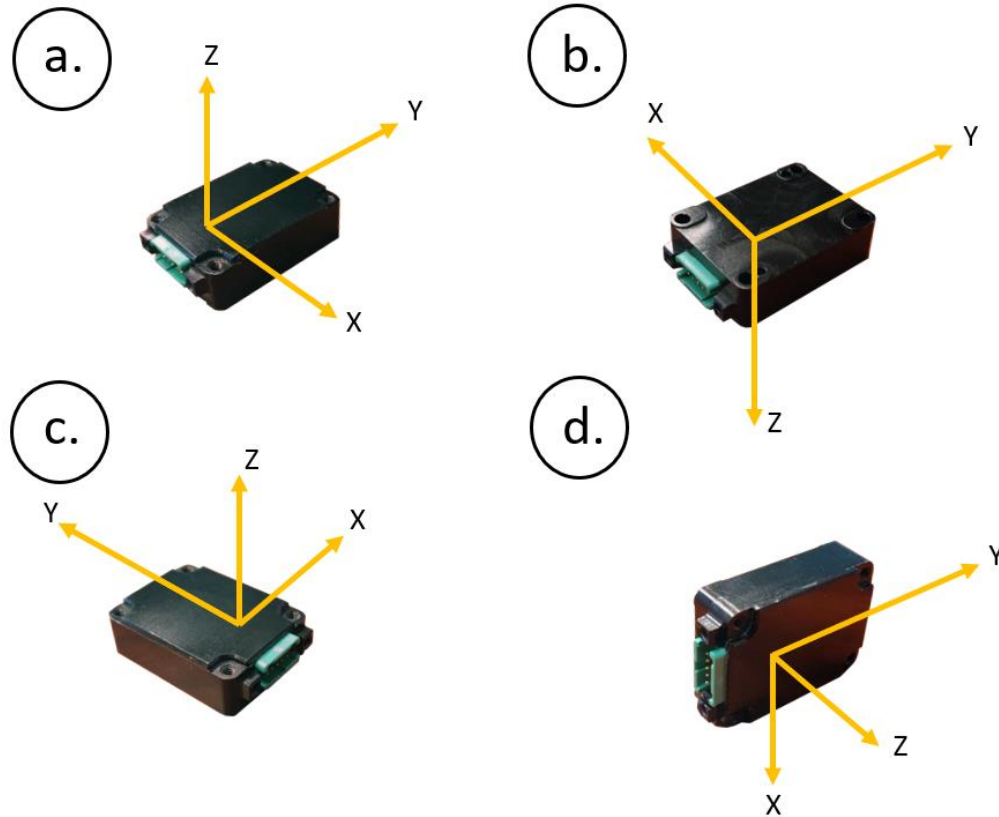


Figure C.1. Examples of the Inertial Labs™ KERNEL (connector location relative to device axis is same as KERNEL) mounting on the carrier object

a – alignment angles are	0	0	0	(degrees);
b – alignment angles are	0	0	180	(degrees);
c – alignment angles are	-90	0	0	(degrees);
d – alignment angles are	0	0	90	(degrees).

Appendix D. Forms of the IMU Orientation Presentation

The coordinate system is defined as $Ox_0y_0z_0$ fixed to the carrier object, where Ox_0 axis is lateral and directed to the right; Oy_0 axis is longitudinal and directed forward; Oz_0 axis is normal and directed vertical. At usual installation of the IMU on the carrier object, the IMU appropriate axes should be parallel to the axes as Figure 2.1 illustrates. Additionally, it is possible to install the IMU in any known position relative to the object with known alignment angles (see Appendix C for details).

The Inertial Labs IMU calculates orientation of the coordinate system $Ox_0y_0z_0$ fixed to the carrier object with respect to Cartesian geographical reference frame $Oxyz$ where axes Ox and Oy are in the level and directed to the east and north, and Oz axis is directed up. Such reference frame is also known as ENU (East-North-Up) Earth-level frame.

Measured angles are the standard Euler angles of rotation from the Earth-level frame to the object frame: heading K is first, then pitch θ , and then roll γ (see Figure D.1).

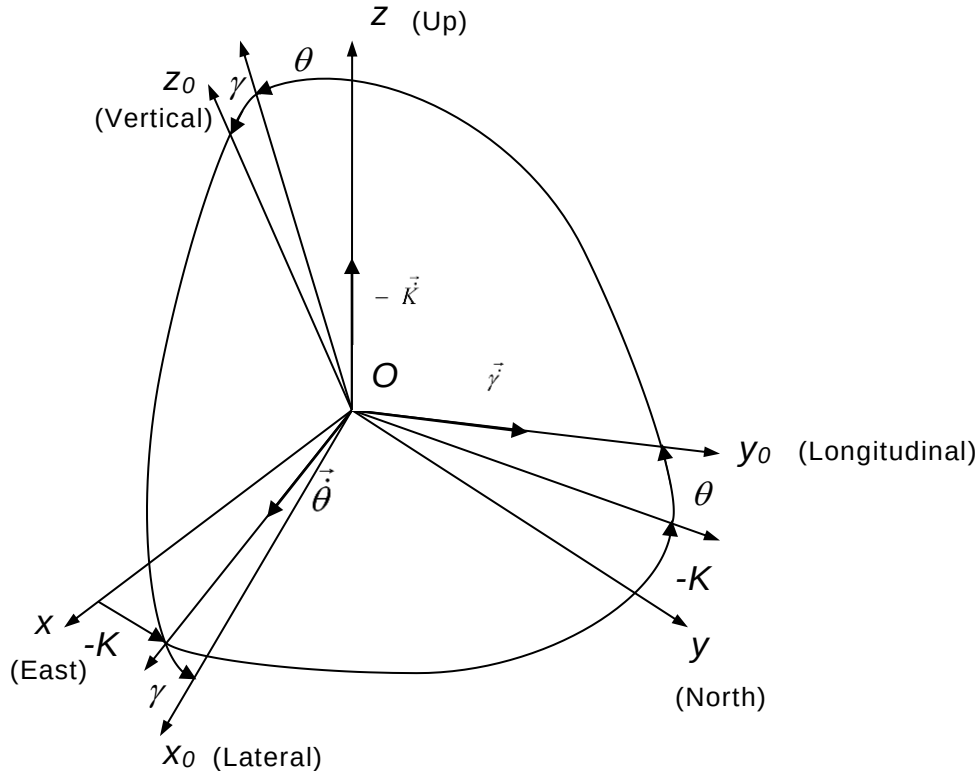


Figure D.1. Transformation of coordinate systems

Notes:

1. Positive direction of heading is clockwise. So, heading K is shown with a minus sign in Figure D.1.
2. In different applications, “heading” is also known as “azimuth” or “yaw”; “pitch” is also known as “elevation” or “tilt”; “roll” is also known as “bank”.

Due to the definition of Euler angles, there is a mathematical singularity when the object longitudinal y_o -axis is pointed up or down (i.e., pitch approaches $\pm 90^\circ$). This singularity is not present in the quaternion or directional cosine matrix (rotation matrix) presentation.

Directional cosine matrix (DCM) is the rotation matrix **C** from the object body reference frame $Ox_o y_o z_o$ to the geographical reference frame $Oxyz$. According to Figure D.1, DCM can be represented through Euler angles as

$$\mathbf{C} = \begin{bmatrix} \cos K \cos \gamma + \sin K \sin \gamma \sin \theta & \sin K \cos \theta & \cos K \sin \gamma - \sin K \cos \gamma \sin \theta \\ -\sin K \cos \gamma + \cos K \sin \gamma \sin \theta & \cos K \cos \theta & -\sin K \sin \gamma - \cos K \cos \gamma \sin \theta \\ -\cos \theta \sin \gamma & \sin \theta & \cos \theta \cos \gamma \end{bmatrix}. \quad (\text{D.1})$$

Or, Euler angles can be calculated from elements c_{ij} of directional cosine matrix **C**:

$$K = \arctan \frac{c_{12}}{c_{22}}; \quad \theta = \arcsin c_{32}; \quad \gamma = -\arctan \frac{c_{31}}{c_{33}}. \quad (\text{D.2})$$

Additionally, the Inertial Labs IMU provides orientation output in quaternion (**Q**) form, which is a hyper-complex number with four components

$$\mathbf{Q} = (q_0, q_1, q_2, q_3), \quad (\text{D.3})$$

where q_0 is real part, q_1, q_2, q_3 are vector part. In other words, q_0 represents the magnitude of the rotation, while the other three components represent the axis about which that rotation takes place. With only four components, quaternion representation of orientation is computationally efficient. However, manipulation of quaternions is not intuitive, so their use in place of directional cosine matrices may increase the chances of unintentional user errors.

Quaternion **Q** is converted to directional cosine matrix **C** using the following expressions:

$$\mathbf{C} = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & q_0^2 + q_2^2 - q_1^2 - q_3^2 & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & q_0^2 + q_3^2 - q_1^2 - q_2^2 \end{bmatrix}. \quad (\text{D.4})$$

The reverse conversation from directional cosine matrix **C** to quaternion **Q** is as follows:

$$q_0 = \frac{1}{2} \sqrt{1 + c_{11} + c_{22} + c_{33}}; \quad (D.5)$$

$$q_1 = \frac{c_{32} - c_{23}}{4q_0}; \quad q_2 = \frac{c_{13} - c_{31}}{4q_0}; \quad q_3 = \frac{c_{21} - c_{12}}{4q_0}.$$

Expressions (D.5) are widely used, but they have singularity at $q_0 = 0$. Therefore, the Inertial Labs IMU uses other expressions that have no singularity:

$$q_0 = \frac{1}{2} \sqrt{1 + c_{11} + c_{22} + c_{33}}; \quad q_1 = \frac{1}{2} \sqrt{1 + c_{11} - c_{22} - c_{33}} \cdot \text{sign}(c_{32} - c_{23}); \quad (D.6)$$

$$q_2 = \frac{1}{2} \sqrt{1 - c_{11} + c_{22} - c_{33}} \cdot \text{sign}(c_{13} - c_{31}); \quad q_3 = \frac{1}{2} \sqrt{1 - c_{11} - c_{22} + c_{33}} \cdot \text{sign}(c_{21} - c_{12}).$$

When needing to calculate Euler angles from quaternion, calculate elements c_{12} , c_{22} , c_{31} , c_{32} , c_{33} , according to (D.6), and then use formulas (D.2):

$$K = \arctan \frac{2(q_1 q_2 - q_0 q_3)}{q_0^2 + q_2^2 - q_1^2 - q_3^2}; \quad \theta = \arcsin(2q_2 q_3 + 2q_0 q_1); \quad (D.7)$$

$$\gamma = -\arctan \frac{2(q_1 q_3 - q_0 q_2)}{q_0^2 + q_3^2 - q_1^2 - q_2^2}$$

where arctan is four-quadrant inverse tangent.

REVISION HISTORY

Revision	Date	Author	Affected Paragraphs	Remarks
1.40	Aug. 28, 2024	KE	All	1. Corrected notes of KERNEL-201 compatibility.
1.39	Jul. 29, 2024	KE	All	1. Added KERNEL-108 compatibility.
1.38	Jul. 04, 2024	VB	7 6.3.7 Page 2 5.1	1. Added description about the message that is automatically sent while the device starts in Auto Start mode. 2. Corrected SaveFlash command example #3. 3. Added the contact information table. 4. Specified that the data latency value depends on the KERNEL type and can be seen in the appropriate datasheet.
1.37	Apr. 18, 2024	KE	All	1. Added KERNEL-102 and KERNEL-201 compatibility.
1.36	Mar. 12, 2024	KE	5.1	Corrected notes on 1PPS and external trigger.
1.35	Feb. 07, 2024	VB	6.3.7, 6.3.8 6.3.7	1. Added description of the 'sensors measurement range' parameter (code 0x8F 0x02), which enables retrieving the measurement ranges of the installed gyroscopes and accelerometers in the device. 2. Specified parameters which are not supported in KERNEL device.
1.34	Jan. 18, 2024	VB	6.5	Updated the values in KERNEL maximum data rate for different output data formats table.
1.33	Dec. 05, 2023	VB	All 6.3.9 6.3.8 Page 2 End of document	1. Removed the LoadPar and ReadPar commands from document. 2. Added description of ReadRam command. 3. Added and described the new parameters that can be changed by the user through the direct command. 4. Arranged the disclaimer and trademark legal notice at the beginning of the document. 5. Arranged the document revision history at the end of the document.
1.32	Oct. 19, 2023	VB	6.2.1 – 6.2.6	Removed information about input voltage (V _{in} p).
1.31	Aug. 31, 2023	DB	6.2.8	Available in the units firmware since August 2023: Added the “Second Fraction” data type (code 0x05) for output through UDD. This option is applicable to all KERNEL IMUs except KERNEL-100.
1.30	Aug. 08, 2023	MJ	5.4	1PPS availability note added.
1.29	Aug. 02, 2023	MJ	5.4	Revised Table 5.3 (signal descriptions corrected).
1.28	Jul. 13, 2023	DB	2, 4.3, 5.4 All All	1. Added preliminary information about the new IMU type: KERNEL-110M. 2. Renamed the LoadKernelPar command to LoadPar.

				3. Renamed the ReadKernelPar command to ReadPar.
1.27	May. 10, 2023	DB	5.1	1. Specified that the 1PPS input is available only if the IMU has firmware released in May 2023 or later. 2. Removed notes that TOV is available by request, this limitation is not actual. The TOV output is supported in IMUs by default.
1.26	May. 02, 2023	DB	5.1 6.2.5 6.5	1. Added description of pin #3 (1PPS) on the main connector and added note on how to use it (see Table 5.1). 2. Corrected Gyros and Accels scale factors in the KERNEL Calibrated HR Data format description (see Table 6.9). 3. The notes regarding the device baud rate setting in Table 6.9 have been duplicated in section 6.5. 4. To the list of supported baud rates added the value of 4000000. It is designed for specific purposes only and requires a particular adaptor for data/commands transmission.
1.25	Apr. 24, 2023	DB	4.1 4.2 5.1 6.3.9 Appendix C 2	1. Added note about a surface finish for the KERNEL-100. 2. Revised notes under Figure 4.3. 3. Revised the description of pin #2 (BOOT) on the main connector in Table 5.1 and added notes on how to use it. 4. Corrected the baud rate parameter length from "0x02 0x00" to "0x01 0x00" (see Table 6.25). 5. Rewritten the alignment angles description for clarity. 6. Updated the KERNEL IMU coordinate system picture (see Figure 2.1).
1.24	Jan. 10, 2023	DB	4.2 4.2	1. Added note with weight information for KERNEL-110, KERNEL-111, KERNEL-120, KERNEL-210, and KERNEL-220 models. 2. Updated the mechanical drawing of the KERNEL-110, KERNEL-111, KERNEL-120, KERNEL-210, and KERNEL-220 models (see Figure 4.3).
1.23	Dec. 12, 2022	DB	All 6.3.9 6.2.2, 6.2.5 6.2.8 All	1. Added preliminary information about the new IMU type: KERNEL-111. 2. Added description to the SaveFlash command with appropriate examples. 3. Revised description of the Counter bytes within the output data payloads. 4. Revised description to User Defined Data options in Table 6.14. 5. Removed information related to the magnetic data since the IMU does not measure the magnetic field value.

			6.5 6.3.3, 6.3.4, 6.3.5, 6.3.6 Appendix C	6. Removed section 6.5 (“Change of the IMU COM Port Baud Rate”) as an uninformative one. The necessary information can be found in section 5.3.9 (“SaveFlash Command”). 7. Revised description to the LoadKERNELPar, ReadKERNELPar, GetDevInfo, and GetBIT commands. 8. Revised description of the IMU alignment angles and specified their ranges.
1.22	Aug. 30, 2022	DB	5.1	Revised supply voltage ranges in Table 5.2.
1.21	Jun. 29, 2022	DB	6.9	Added section “6.9. Cascade Filter”.
1.20	Jun. 27, 2022	DB	All	Added preliminary information about the new IMU types: KERNEL-210 and KERNEL-220.
1.19	Mar. 29, 2022	DB	4.3 4.3.1 6.2.8	1. Updated outline drawing for the KERNEL-110 and KERNEL-120. 2. Added section “4.3.1. Center of Navigation”. 3. Added the following data types to the UDD list: · Aux accelerometer (id 0x2F) · Filtered low-G accels HR (id 0x91) Filtered high-G accels HR (id 0x92).
1.18	Feb. 09, 2022	DB	6.1 6.2.8 6.2.8 6.5 6.6 6.8	1. Added section “6.1. Operational Modes”. 2. Added data type 0x2E (Gyro Bias) to UDD list. 3. Specified that the User Defined Data format structure corresponds to the common byte structure for all commands and messages to/from the KERNEL IMU unit. 4. Added important note about the 1843200 and 3686400 baud rate usage. 5. Revised Table 6.25. 6. Added section “6.8. Initial alignment logic”.
1.17	Feb. 04, 2022	DB	6.4 6.2.1 All All 6.1.6, 6.1.7, 6.2, Appendix B Appendix B Appendix D All All	1. Added section “6.4. Change of the IMU COM port baud rate”. 2. Corrected code of KERNEL_QuatData command in Table 6.16 from 0x36 on 0x82. 3. Removed duplicate tables with KG and KA factors. 4. Added preliminary information about the new IMU types: KERNEL-110 and KERNEL-120. 5. Added description of KERNEL GAA and KERNEL GAAM data formats. Also added appropriate KERNEL_GAAMData and KERNEL_GAAMData commands. 6. Corrected the command structure for the KERNEL_GAMData command in Table B.1. 7. Added “Appendix D. Forms of the IMU Orientation Presentation”. 8. Revised the contact information at the footer. 9. Added disclaimer and trademark legal notice at the end of the document.
1.16	Oct. 13, 2021	DB	6.1	Removed the “Check-Sum” bytes from payload to each data format.

1.15	Aug. 20, 2021	AA	6.1.6 6.3	1. Added data types 0x2D and 0x2B to UDD list. 2. Added description of USW Extended
1.14	Jun. 23, 2021	DB	All 5.1 6	1. The document general layout improvements. 2. Corrected note #3 (under Table 5.2) about TOV behavior. 3. Corrected important note under Table 6.3, where specified that in all multi-byte data LSB is transmitted first.
Revision	Date	Affected paragraphs	Remarks	
1.13	May 3, 2021	Section 6.2.8	Adding Section 6.2.8 “Changing the IMU Output Variant Type” and including message description and definitions.	
1.12	15.Mar.2021	Section 4.1 Section 5.4 Section 6 Section 6.5 Section 6.6	Updated Initial Alignment time from 3 seconds to 2; Updated default baud rate to 2000000; Created Section 6.5 “FIR Filter Delay”; Created Section 6.6 “KERNEL-100 Automatic Start”; Update Figure 4.1; Updated note under TOV description.	
1.11	17.Feb.2021	Section 4	Updated Mechanical Drawing	
1.10	04.Feb.2021	Section 5, 6	Updated Table 6.25; updated mechanical drawing.	
1.9		Section 6.1.6 Section 5.4 Section 6.1 Section 4	Added Table 6.13 “Payload of the “User Defined Data” Updated the TOV Graphic Added the “Check-Sum” byte to all data formats where missing. Updated Figure 4.1	
1.8	30.Nov.2020	Appendix C, Section 6.3, Section 5.4, Section 5.1.	Updated definition of the heading alignment angle; added multi-interface information; Updated USW definition; Updated information on EXTRIG description; Updated information on TOV description; Updated information in Table 5.2	
1.7	4.Nov.2020	All	Removed PPS information in Section 6.1.3; General update for grammar and readability.	
1.6	15.Oct.2020	Section 5, Section 6.1.6, Section 6.2.8, Section 6.2.7	Update voltage from Table 5.1, Table 5.2 – delete line termination string and update voltage range, added User Defined Data. Removed Initial Alignment payload information. Modified Section A (RS-232 option); updated links on page 30 and 32; updated Figure C.1; removed mating option 1 from pages 6 and 7.	
1.5	13.Oct.2020	Section 6	Added Definition for KA/KG for all formats; updated 1864 gyro to be 2000 gyro.	
1.4	23.Sept.2020	Section 6.1.5	Added Calibrated HR Data as Available Data Format	
1.3	22.Sept.2020	Appendix A	Corrected the Code and Exact Structure of the QuatData Command	
1.2	15.Sept.2020	Sec. 6.1.4	Added Section for KERNEL Quaternion Data format.	

1.1	14.Sept.2020	6.1.1	Calibrated changed to Orientation data.
1.0	8.Sept.2020	All	Initial Release

